Paleomagnetic study of the late Neoproterozoic Bull Arm and Crown Hill formations (Musgravetown Group) of eastern Newfoundland: implications for Avalonia and West Gondwana paleogeography[1]

Sergei A. Pisarevsky, Phil J.A. McCausland, Joseph P. Hodych, Sean J. O'Brien, Jennifer A. Tait, and J. Brendan Murphy

Abstract: A paleomagnetic study of subaerial volcanic rocks and associated siltstones of the Ediacaran Bull Arm Formation in the Avalon Zone of Newfoundland revealed a stable bipolar, hematite-borne primary remanence supported by positive conglomerate, contact, and fold tests. Mean remanence directions in two distal areas (Bonavista and Argentia) are similar, indicating a low paleolatitude position of Avalonia at ~570 Ma. Redbeds of the overlying ~550 Ma Crown Hill Formation also carry a primary bipolar hematite-borne remanence with moderate inclination, indicating that Avalonia remained at low to medium paleolatitudes through the end of the Ediacaran. Combining our results with previously published paleomagnetic data of Avalonia suggests moderate-scale drift of Avalonia through low southern paleolatitudes through the latter half of the Ediacaran, providing a paleogeographic context for the development of the first complex metazoan life.

Résumé : Une étude paléomagnétique de roches volcaniques subaériennes et de siltstones associés de la Formation de Bull Arm (Édiacarien) de la zone Avalon de Terre-Neuve révèle une bipolarité stable, une rémanence primaire dans l'hématite, laquelle est supportée par un conglomérat positif, un contact et des études sur les plis. Les directions moyennes de la rémanence dans deux secteurs distaux (Bonavista et Argentia) sont semblables, indiquant une basse position paléolatitude d'Avalonia vers ~570 Ma. Les lits rouges de la Formation de Crown Hill (~550 Ma) sus-jacente conservent aussi une rémanence primaire bipolaire dans l'hématite avec une inclinaison modérée, indiquant qu'Avalonia est demeuré à des paléolatitudes basses à moyennes jusqu'à la fin de l'Édiacarien. Nos résultats combinés à des données publiées antérieurement sur Avalonia suggèrent un déplacement à échelle moyenne d'Avalonia à travers de basses paléolatitudes sud durant la deuxième moitié de l'Édiacarien, fournissant un contexte paléogéographique pour le développement de la première vie complexe des métazoaires.

[Traduit par la Rédaction]

Introduction

The late Neoproterozoic was a time of dramatic paleogeographic changes. One supercontinent (Rodinia) finally broke up and another one (Gondwana) began to assemble. Several oceans (Mozambique, Adamastor, Brazilliano) closed during the formation of Gondwana as others (Iapetus, Paleo-Asian, Tornquist) began opening between Laurentia, Baltica, Siberia, Gondwana and other cratonic blocks. However, the paleogeographic reconstructions for this time remain controversial (e.g., Dalziel 1997; Cordani et al. 2003; Meert and Torsvik 2003; Tohver et al. 2006; McCausland et al. 2007, 2011; Li et al. 2008; Pisarevsky et al. 2008).

The main reason for this controversy is the sparsely populated Neoproterozoic paleomagnetic database (McElhinny

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and Lock 1996; Pisarevsky 2005). Very few reliable and well-dated paleopoles are available for the major cratons through the late Neoproterozoic (see reviews of Collins and Pisarevsky 2005; McCausland et al. 2007; Pisarevsky et al. 2008). Consequently, these data do not allow unequivocal global paleogeographic reconstructions for the time interval between ~620 and 540 Ma (the Ediacaran Period; Knoll et al. 2004). There is still a paucity of reliable Ediacaran paleomagnetic data from many major cratons, such as West Africa, Congo, Kalahari, Siberia, Amazonia, and others. There is also controversy regarding interpretation of Ediacaran paleomagnetic data from Laurentia and Baltica (McCausland and Hodych 1998; Pisarevsky et al. 2000, 2001; Meert and Van der Voo 2001; Cawood and Pisarevsky 2006; McCausland et al. 2007, 2011).

Paleomagnetic poles from smaller terranes are rarely used for global paleogeographic reconstructions. Block rotations, oroclinal bending, and other regional and local processes occur during the accretion of small terranes to large continents and their subsequent transport along plate boundaries (e.g., Coney et al. 1980; Abrajevitch et al. 2008). The exact orientations of small blocks at different times during these processes are, therefore, hard to reconstruct. Consequently, only the paleolatitudes obtained from paleomagnetic studies of small terranes are usually applicable as constraints for the paleopositions of paleocontinents.

Although faunal endemism indicates temporary separation of Avalonia (parts of Newfoundland, Nova Scotia, New England, southern UK, and Belgium) from Gondwana in the late Ediacaran and Early Cambrian (Landing 2005), Avalonia is usually considered to have been located among the Peri-Gondwanan terranes that were distributed near the northern (present coordinates) margin of Gondwana, adjacent to Amazonia and (or) West Africa at ~600–540 Ma (e.g., Kroner and Hegner 1998; Friedl et al. 2000; McNamara et al. 2001; Murphy et al. 2004; Thompson et al. 2007; Nance et al. 2008, 2010; Pisarevsky et al. 2008 and references therein). Avalonia's paleogeographic position is particularly important because regional tectonic studies indicate that it faced an open ocean from the late Neoproterozoic to the early Paleozoic (Murphy and Nance 1989).

The first paleomagnetic data from West Avalonia were published more that 50 years ago (Nairn et al. 1959). However, most published results cannot be considered reliable because of a lack of convincing evidence for a primary remanence and (or) poor geochronological control. Only a few paleomagnetic results pass these modern criteria, as will be discussed later in the text. In this work, we present a new paleomagnetic study of mid to late Ediacaran volcanic rocks and redbeds of the Bonavista and Avalon peninsulas, Newfoundland, and discuss their implications for understanding the paleogeographic history of Avalonia with respect to Gondwana at the end of the Neoproterozoic.

Geology and sampling

The Bonavista and Avalon peninsulas (Figs. 1, 2) are parts of the Avalonian tectonostratigraphic zone of Newfoundland (Williams 1979), which represents the core portion of West Avalonia (O'Brien et al. 1996). Paleomagnetic sampling was conducted during the summer of 2007 in two localities: (1) in the northern portion of the Bonavista peninsula in the vicinity of Plate Cove West, Red Cliff and King's Cove, and (2) on the western shore of the Avalon peninsula, near Argentia. The latter sampling supplements a pilot study conducted by Jordan (1999), the results of which are also included herein.

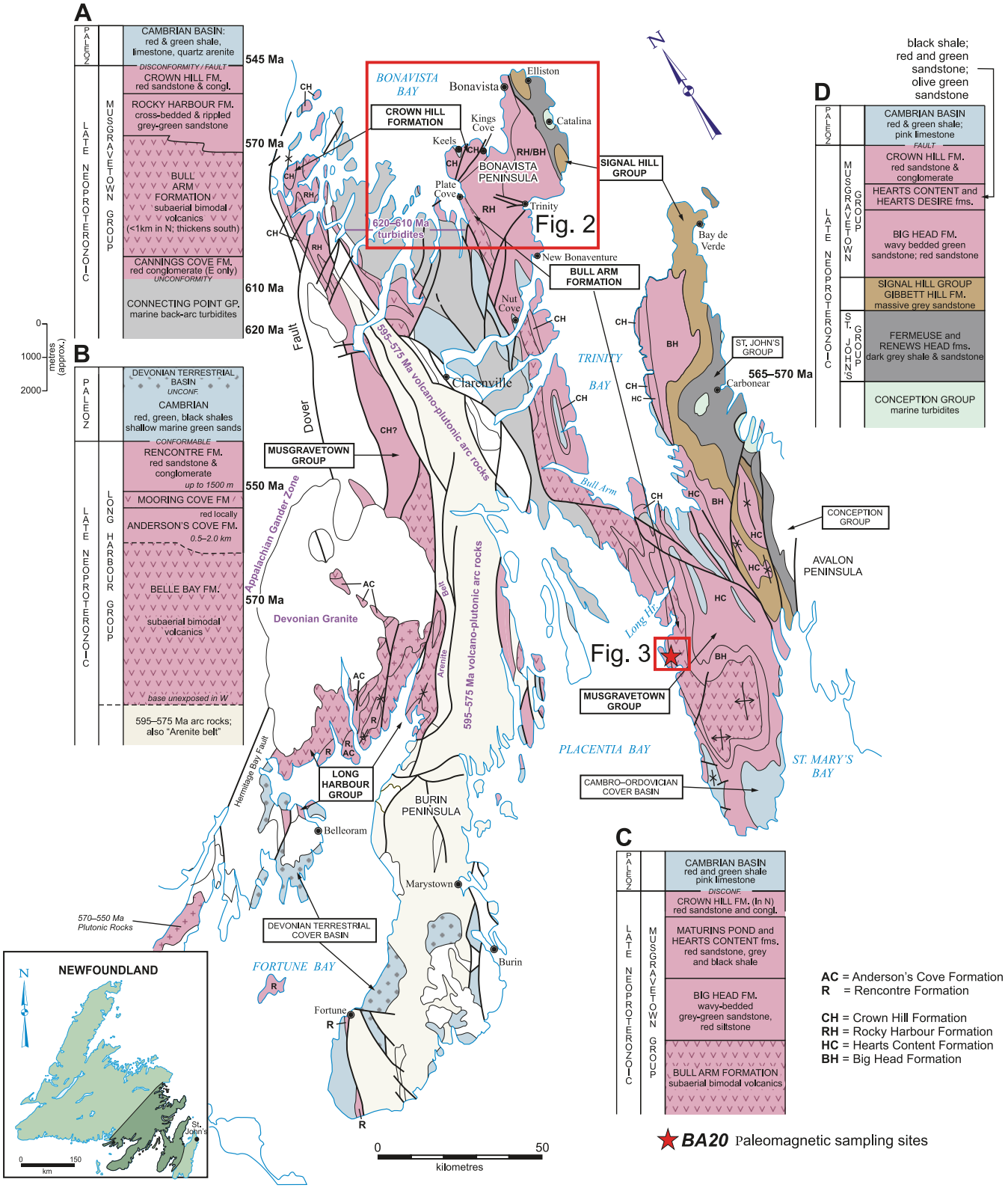
Bonavista locality: Bull Arm and Crown Hill formations

Neoproterozoic rocks in the eastern part of the Bonavista peninsula belong to two Neoproterozoic sedimentary basins: (1) the eastern St. John's basin (Figs. 1D, 2), which includes the Conception, St John's, and Signal Hill (Gibbett Hill Formation) groups; and (2) the western Bonavista basin (Figs. 1A; 2), which includes the Connecting Point and Musgravetown groups (O'Brien and King 2002; Normore 2010). The basins are separated by the Spillars Cove – English Harbour fault (Fig. 2; Normore 2010). All these strata were folded and faulted in the Siluro-Devonian during collision of Avalonia with Laurentia (Dallmeyer et al. 1983; O'Brien et al. 1996, O'Brien and King 2002). Oriented paleomagnetic samples were collected from two units of the Musgravetown group: bimodal subaerial volcanic rocks of the Bull Arm Formation and younger redbeds of the Crown Hill Formation (Figs. 1, 2).

The Bull Arm Formation consists mainly of dark vesicular basalt of volcanic-arc affinity with synvolcanic epidote and hematite alteration, intercalated with reddish massive rhyolites and subordinate clastic sedimentary rocks (O'Brien and King 2002; Normore 2010). The subaerial to marine, typically bimodal succession can be traced southwards from the Plate Cove area of the Bonavista peninsula through the type locality at Bull Arm to the Placentia Bay coast of the Avalon peninsula (Fig. 1); it also crops out on the western coast of Bonavista Bay (O'Brien and Knight 1988; O'Brien and King 2002). The Bull Arm Formation overlies red conglomerate (Cannings Cove Formation, Fig. 1A) of the basal Mugravetown Group, which in turn lies with pronounced angular discordance on the 620–610 Ma turbidites of the Connecting Point Group (Jenness 1963; Dec et al. 1992; O'Brien 1993; Fig. 1). The Bull Arm Formation passes upward into the Rocky Harbour Formation (Fig. 1A) — a widespread shallow-marine clastic succession. In the western part of Bonavista Bay, the Bull Arm Formation at its contact with the overlying Rocky Harbour Formation has been dated at $570\ ^{+5}_{-3}$ Ma (U–Pb zircon, O'Brien et al. 1989), providing a younger limit to the age of the Bull Arm magmatism in that part of the Avalon Zone. The sampled Bull Arm Formation lavas from the Bonavista peninsula (Fig. 2) are correlated on a regional scale with latter units and are, thus, assumed to be coeval (within analytical error limits).

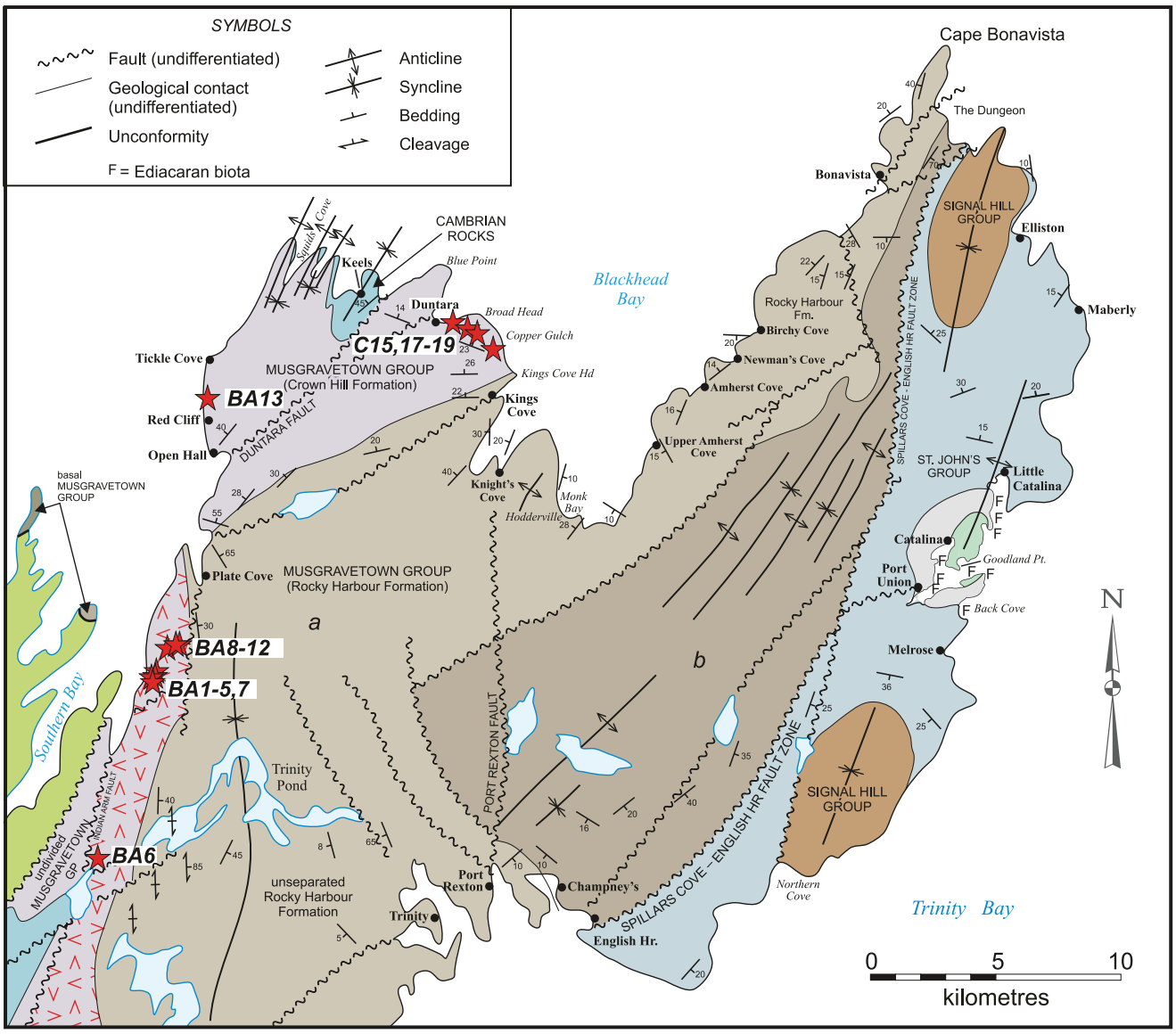
The main sampling locality for the Bull Arm Formation lies along highway roadcuts and adjacent quarries between Summerville and Plate Cove West, with one other quarry location on Highway 230, 6 km to the south of Summerville (Fig. 2). Along a large cliff-forming roadcut and in hilltop quarries, successive flows of the Bull Arm Formation are well exposed, dipping moderately to steeply eastwards. In these sites, we were able to establish a stratigraphic order within which to interpret the paleomagnetic results. Paleohorizontal for some sites was not easy to determine, given the massive appearance of some flows and only occasional occurrence of stratified siltstone. We have applied tilt correction

Fig. 1. Simplified map of the Avalon Zone, Newfoundland (simplified after O'Brien and King, 2005). A–D, areal stratigraphic columns. Locations of Figs. 2 and 3 are shown.



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Fig. 2. Simplified map showing paleomagnetic sampling sites and regional geological elements of Bonavista peninsula, Newfoundland (after O'Brien and King 2005).



EARLY PALEOZOIC
CAMBRIAN

Unseparated Random, Bonavista, and Smith Point formations: quartz arenite, red and green shales, limestone and slate

LATE NEOPROTEROZOIC
EDIACARAN

MUSGRAVETOWN GROUP

Crown Hill Formation
mainly red sandstone and conglomerate; minor grey sandstone, siltstone and red mudstone; variegated tuff and sandstone

a | b Rocky Harbour Formation
a: unseparated grey, green and locally red cross-bedded and rippled sandstone; pebble to cobble conglomerate;
b: siliceous grey-green siltstone

Bull Arm Formation
unseparated subaerial volcanic flows and pyroclastic rocks; minor clastic sedimentary rocks

BA13 Paleomagnetic sampling sites

SIGNAL HILL GROUP

Gibbett Hill Formation
massive to laminated light grey sandstone

ST. JOHN'S GROUP

unseparated Fermeuse and Renews Head formations: mainly grey to black, locally fossiliferous* and variably pyritic sandstone and shale; unseparated slumped units

Trepassey Formation: locally fossiliferous* green argillite and sandstone

CONCEPTION GROUP
unseparated Drook and Mistaken Point formations: silicified, graded grey and green sandstone and mudstone; fossiliferous* grey, green and minor red argillite

CONNECTING POINT GROUP (west of Indian Arm Fault)

mainly unseparated grey-green turbidites and associated marine sedimentary rocks

*Ediacaran biota

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locally where possible, but in some cases have extrapolated paleohorizontal over tens of metres of stratigraphic section. Separate cooling units were sampled by recognizing lithological breaks across siltstone units, alternation between mafic and felsic flows or the occurrence of ferruginous oxidized flow tops. In all, 11 separate flows from the Bull Arm Formation were sampled in the Bonavista locality.

The overlying Rocky Harbour Formation contains conglomerates with mafic and felsic cobbles derived from the Bull Arm Formation. A large (2 m × 2 m × 1 m) boulder of conglomerate was sampled next to the Bull Arm sample sites in the Plate Cove area, to perform a paleomagnetic conglomerate test (Graham 1949) of the underlying Bull Arm Formation.

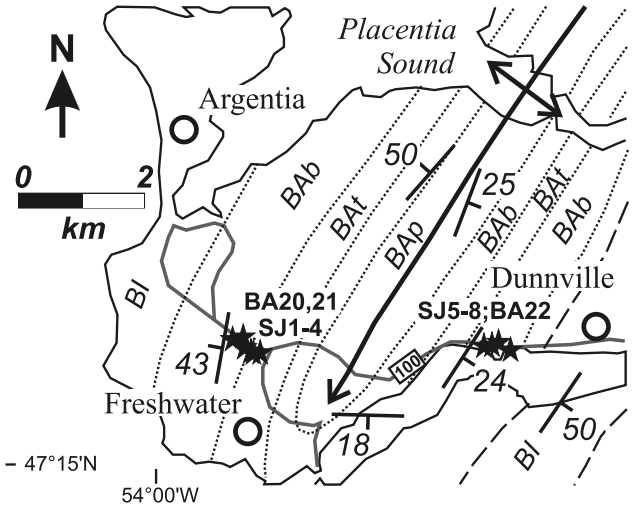
The Crown Hill redbeds conformably and gradationally overlie the Rocky Harbour Formation (Fig. 1A) and outcrop in the northern part of the western horn of the Bonavista peninsula (Fig. 2). We collected red siltstones to mudstones from gently northeast-dipping beds at shoreline exposures in the Duntara area and in three roadcut outcrops along the Duntara – King's Cove Highway (sites C15–C19; Fig. 2). Red conglomerate of the Crown Hill Formation exposed near Red Cliff contains mafic and felsic volcanic pebbles of the Bull Arm Formation (Fig. 2); these were sampled for a second, regional conglomerate test (site BA13). The Crown Hill redbeds are disconformably overlain by Lower Cambrian red and green shales and minor limestones of the Bonavista Formation, exposed in the core of an open syncline at Keels that plunges gently to the north (Fig. 2). Penetrative deformation is pervasive in the core of the syncline, but it is represented only by widely spaced, subvertical cleavage planes in the Crown Hill Formation redbeds at Duntara, east of the Duntara fault. Sites C15–C19 in the Crown Hill redbeds between Duntara and King's Cove do not appear to require plunge correction as they lie within a structure that is separate from the Keels syncline.

The youngest detrital zircons in the Crown Hill Formation are ~557 Ma (Pollock et al. 2009). Hence, the age of the studied redbeds is bracketed between 557 and 542 Ma, the age of the base of the lower Cambrian. The Crown Hill Formation has been correlated with similar latest Neoproterozic redbeds of the Rencontre Formation (Long Harbour Group), exposed in the southwestern Avalon Zone (O'Brien et al. 1990, 1995; O'Brien and King 2002). These redbeds overlie rhyolites dated at 551 ± 6 Ma (O'Brien et al. 1995; Tucker and McKerrow 1995). This regional correlation is consistent with the detrital zircon results and makes the depositional age constraints for Crown Hill slightly tighter: 551–542 Ma.

Argentia locality: Bull Arm Formation

For regional comparison the Bull Arm Formation volcanic rocks were also sampled in the Argentia area of the southwestern Avalon peninsula (Fig. 1, 3), which follows up on the pilot study by Jordan (1999). Here, the age constrains are less certain — the Bull Arm Formation in Argentia is thought to be correlative with the Bonavista exposures (e.g., ca. 570 Ma), although no geochronological data exist from this area. Regional tectonism of the western Avalon Zone of Newfoundland, which includes both Argentia and Bonavista areas, has been correlated with that of the adjoining Gander Zone and is the result of the juxtaposition of the zones during regional sinistral transpression along the Dover Fault shear zone (Fig. 1). Deformation and synkinematic magmatism along this crustal boundary is coeval with orogen-scale tectonism and is dated as mid- to late Silurian (Dunning et al. 1990, Holdsworth 1991, Schofield and D'Lemos 2000; O'Brien et al. 1996; van Staal et al. 1998, 2009; Murphy and Nance 2002).

Fig. 3. Simplified map showing paleomagnetic sampling sites and regional geological elements of Argentia area, Newfoundland.



Ten paleomagnetic sites were collected by oriented block sampling (Jordan 1999) and with a portable drill (in this study), from gently west- and east-dipping flows and intercalated siltstones on both limbs of an open anticline exposed in roadcuts along Highway 100 west of Dunville (Fig. 3; King 1988). Paleohorizontal was readily obtained from the intercalated siltstones. From regional mapping, the sample sites in the east limb are approximately at the same stratigraphic level as those of the west limb (King 1988), allowing for a simple paleomagnetic fold test (Graham 1949).

Analytical methods

Remanence behaviour was determined by detailed stepwise thermal demagnetisation (≤20 steps, to 600 °C), using a Magnetic Measurements MMTD1 furnace and the 2G cryogenic magnetometer in the University of Edinburgh (Edinburgh, Scotland). Stepwise alternating field (AF) demagnetisation (≤20 steps, up to 100 mT) was also applied, using an AGICO LDA-3A tumbling demagnetiser. Magnetic mineralogy was investigated from demagnetisation characteristics and, in selected samples, from measuring the variation of susceptibility versus temperature (20–700 °C) using the AGICO Kappabridge MFK-FA with the CS3 apparatus. Parts of the collection were studied in the paleomagnetic laboratories of Oxford University (Oxford, UK), Utrecht University (Utrecht, The Netherlands), The University of Western Australia (Perth, Australia), and University of Windsor (Windsor, Ontario, Canada). The Argentia locality pilot study (Jordan 1999) was undertaken at Memorial University (St. John's, Newfoundland, Canada), with experimental procedures simi-

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lar to those of McCausland and Hodych (1998). Magnetisation vectors were isolated using Principal Component Analysis (Kirschvink 1980).

Paleomagnetic analysis

Bull Arm Formation, Bonavista locality

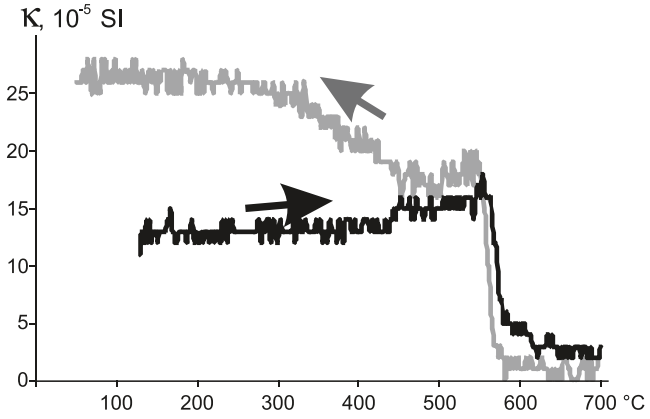
Petrographic analysis indicates that the mafic flows of the Bull Arm Formation are variable in grain size and texture. The finer grained samples contain rare microphenocrysts of plagioclase and pseudomorphs of clinopyroxene set in a fine-grained matrix dominated by chlorite and microlites of plagioclase that display eutaxitic flow textures. Both the phenocrysts and the microlites are extensively saussuritized and are primarily altered to epidote, chlorite, and minor calcite. Opaque minerals occur as (*i*) abundant, fine ragged grains homogenously distributed throughout the matrix, (*ii*) inhomogenously distributed clusters that surround clinopyroxene pseudomorphs, (*iii*) fine needles within altered plagioclase, and (*iv*) in localized fractures that occur at variable angles to the flow fabric. For each occurrence of opaques, magnetite exhibits partial to extensive alteration to hematite.

The coarser grained rocks are less altered and contain microphenocrysts of augite and plagioclase (now albite). The augite is partially pseudomorphed by chlorite and epidote. Plagioclase is variably saussuritized. Opaques are dominated by magnetite, which is partially to extensively altered to hematite. The freshest flows are relatively coarse grained and contain olivine that, although variably altered, preserves some relatively fresh grains. Even in these samples, however, the opaque minerals are predominantly ragged grains that are homogenously distributed throughout the matrix and are moderately to extensively altered to hematite.

Magnetic susceptibility vs. temperature experiments on specimens from the Bull Arm volcanic rocks indicate a presence of both magnetite and hematite as potential remanence carriers (Fig. 4). AF demagnetisation was not effective for the studied rocks — the maximum field of 100 mT was not enough for proper isolation and removal of the characteristic remanent magnetization (ChRM), indicating that the remanence contribution from hematite is dominant. Thermal demagnetisation of mafic rocks revealed two stable remanence components with unblocking temperatures of 500–580 and >630 °C (Fig. 5). The medium temperature component, carried by magnetite, is rather scattered, with the varying of site mean α_{95} between 16° and 52° (Table 1). In situ site mean directions weakly cluster with a south, shallow down direction, best represented by sites BA3 through BA7. The overall mean direction for the mid temperature component is characterized by a Fisher's precision parameter $k < 3$, and is too poorly described to warrant further statistical analysis. Demagnetisation of rhyolites associated with some sites was not successful — the isolated remanence component directions were random and excluded from further analysis.

The high-temperature hematite-borne component in the mafic rocks is significantly less scattered than the low-temperature component, and it is bipolar (Table 2; Fig. 6). If the sites from the main sampling locality north of Summerville are placed in stratigraphic order, then, from the bottom to the top we have a "normal" (SW downward) interval (sites BA5 to BA1), then an "inverse" (NE upward) interval (sites BA8, BA11, and BA10) and again a "normal" interval on the top (site BA12). This apparently regular magnetostratigraphic pattern favours a primary nature of the high-temperature component. The stratigraphic position of the geographically isolated site BA6 ("inverse") with respect to the main section is not clear.

Fig. 4. Magnetic susceptibility (K) versus temperature curves, sample BA10A.

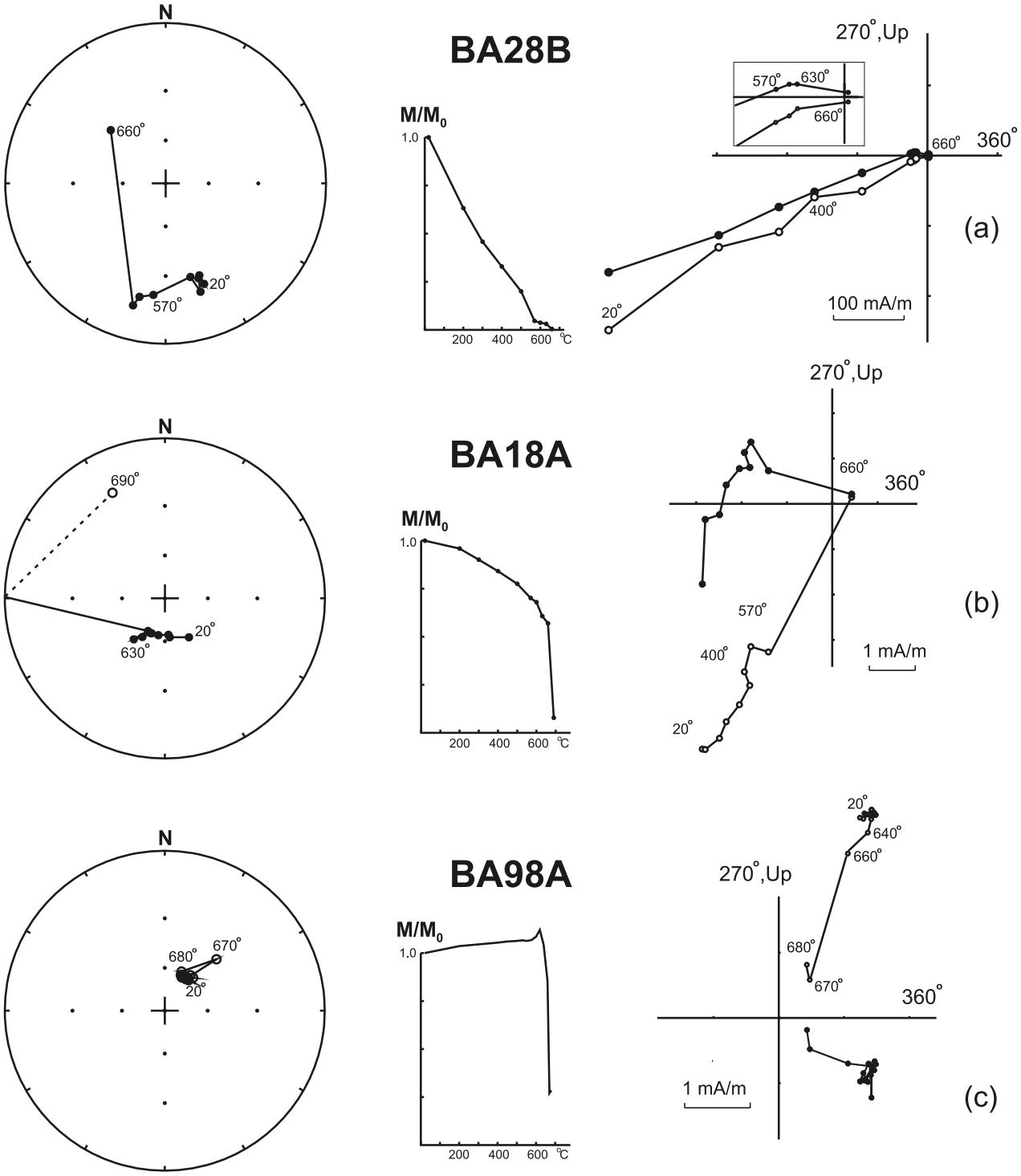


Most sites exhibit a Fisherian distribution of specimen directions (Fisher 1953). One exception is site BA7, which produced an elongated distribution and was excluded from further interpretation (Fig. 7). The mid-temperature and high-temperature mean directions of site BA2 are consistently ~90° clockwise rotated with respect to those of adjacent sites BA1 and BA3 (Tables 1, 2), suggesting that this site has experienced an unrecognized local block rotation with respect to the stratigraphy. It is noteworthy that site BA2 was sampled from a blocky, massive flow that outcropped in a vertical fissure through a high road cut and has no paleohorizontal control.

The nine site-mean directions form two roughly antipodal groups (Fig. 6*a*), but the reversal test of McFadden and McElhinny (1990) is negative. However, if site BA5 is excluded because of its 37° difference from the mean direction, the test becomes positive (Rc, $\Delta = 18.2°$, $\Delta_c = 19.5°$; Fig. 6*b*). As this exclusion might be considered as somewhat biased, we include site BA5 in the overall mean calculation. In our view, the reason for the reversal test uncertainty is the paucity of bedding constraints in the Bonavista section. The stratified sediments associated with the flows are sparingly exposed, so we were forced to extrapolate their bedding; note that the flows and bedding dip moderately to steeply towards the northeast in this section, so uncertainty in paleohorizontal will be manifested as inclination errors in the high-temperature mean directions, causing poor statistics and uncertainty in the reversal test.

Given these circumstances, we interpret the overall tilt-corrected mean direction of declination (D) = 209.8°, inclination (I) = 34.7° (α_{95} = 16.8°), based on five "normal" and four "reversed" site mean directions, to adequately represent the ancient hematite-borne magnetization in the Bull Arm Formation at the Bonavista locality.

Fig. 5. Thermal demagnetisations of the Bull Arm mafic volcanics. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetisation. Stereoplots (equal-angle projection) show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols. (*a*) Specimen BA28B is a fine grey to purple basalt from site BA4, showing remanence predominantly carried by magnetite, with a small, distinct high-temperature component (inset) carried by hematite. (*b*) Specimen BA18A is from an oxidized flow top at site BA3, showing distributed unblocking of remanence carried by magnetite and a significant unblocking of a SW down remanence at high temperature corresponding to hematite. (*c*) Specimen BA98A is from an amygdaloidal basalt flow at site BA11, showing hematite as the only significant carrier of a "reversed" NE and upwards-directed remanence.



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Table 1. Bull Arm Formation mafic rocks, mid-temperature (magnetite) component.

Site	Location		DipD (°)	Dip (°)	*N*/*n*	Dg (°)	Ig (°)	*k*	α_{95} (°)	Ds (°)	Is (°)	*k*	α_{95} (°)
	Latitude (N)	Longitude (W)											
Bonavista locality													
BA1	48°27.07′	53°32.48′	92	66	11/7	268.5	14.8	6.5	25.7	251.6	80.2	6.5	25.7
BA2	48°27.07′	53°32.48′	92	66	9/6	139.6	–3.4	8.2	24.9	165.3	–39.8	8.2	24.9
BA3	48°27.07′	53°32.48′	92	66	9/7	208.9	50.1	3.7	36.4	136.1	31.7	3.7	36.4
BA4	48°27.07′	53°32.48′	94	76	11/11	177.9	31.1	3.7	36.4	153.4	21.5	3.7	36.4
BA5	48°27.07′	53°32.48′	94	76	9/6	214.1	64.7	15.3	17.7	118.1	25.3	15.3	17.7
BA6	48°23.65′	53°33.33′	79–80	45–60	11/9	169.7	15.4	6.2	22.4	159.0	20.7	6.2	22.4
BA7	48°27.07′	53°32.48′	94	76	9/7	176.5	13.2	15.4	15.9	169.4	–3.9	15.4	15.9
BA8	48°27.52′	53°31.68′	94	76	7/4	103.0	49.0	4.8	47.0	87.7	4.0	4.8	47.0
BA10	48°27.77′	53°31.67′	84	64	6/5	37.3	22.2	3.2	51.5	43.4	0.8	3.2	51.5
BA11	48°27.77′	53°31.67′	84	64	6/2	90.6	61.0	—	—	87.2	–2.8	—	—
BA12	48°27.77′	53°31.67′	84	64	6/6	277.9	34.4	3.4	43.1	32.2	75.4	3.4	43.1
Argentia locality													
	West Limb												
SJ1	47°16.15′	53°58.79′	300	38	6/6	330.8	74.2	27.6	13.0	310.2	37.9	27.6	13.0
SJ4	47°16.15′	53°58.79′	304	35	5/4	333.8	51.5	51.5	12.9	323.2	19.4	51.5	12.9
BA21	47°16.13′	53°58.54′	304	35	6/4	15.3	52.4	23.1	19.5	347.2	19.5	23.1	19.5
	East Limb												
BA22	47°16.15′	53°55.68′	123	24	8/5	20.6	52.0	52.1	10.7	43.6	54.6	52.1	10.7

Note: DipD, dip direction of bedding; Dip, dip of bedding; *N*/*n*, number of demagnetized/used samples for site mean calculation; Dg, Ig, remanence declination, inclination in situ (geographic system); Ds, Is, remanence declination, inclination after the tilt correction (stratigraphic system); *k*, Fisher's precision parameter; α_{95}, the semi-angle of the 95% cone of confidence.

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Conglomerate tests

Three outcrops were used for conglomerate tests (Graham 1949) of the Bull Arm volcanic rocks. We collected pebbles and cobbles of both mafic and felsic rocks. However, as the paleomagnetic study of rhyolite flows typically revealed scattered directions, we used only mafic cobbles for the conglomerate tests.

The first outcrop (site BA9) is a 2 m × 2 m × 1 m boulder of conglomerate from the overlying Rocky Harbour Formation (Fig. 1). Cobbles of up to 15–20 cm size are represented by mafic and felsic rocks of the Bull Arm Formation. The high-temperature hematite-borne component (Fig. 8*a*) in the mafic cobbles produced a random distribution, but with consistency within cobbles (Fig. 9*a*). The Rayleigh statistics $r = R/n = 0.312$ compared with $r_c = 0.7$ for five cobbles at 95% confidence level (Mardia 1972), so this test is positive.

The second outcrop (site BA13) represents pebble conglomerates of the even younger Crown Hill Formation near Red Cliff (Fig. 2), which contain pebbles of the Bull Arm volcanic rocks in a red siltstone matrix. This is a regional conglomerate test, applicable to both the Bull Arm volcanic rocks and the host Crown Hill Formation. The high-temperature component (Fig. 8*b*, 8*c*) in mafic pebbles shows a propensity for northerly, upwards directions, but nevertheless is also positive (Fig. 9*b*) with $r = 0.44$ compared with $r_c = 0.56$ for eight pebbles. We note that the conglomeratic red beds at Red Cliff are strongly cleaved along planes that dip steeply to the southeast, parallel to the regional fold axial planes (Fig. 2); it is possible that some pebble mean directions with slightly lower (ca. 630 °C) unblocking temperature carry the signal of a partial overprint that is directed in a north and shallow upwards direction (Fig. 8*c*), acquired during folding and penetrative deformation. This apparent north, shallow up overprint direction is antipodal to the in situ magnetite-borne south, shallow down direction found in sites from the Bonavista Bull Arm volcanic rocks (Table 1). However, most of the pebbles carry a hematite-borne remanence (Fig. 8*b*) apparently without any overprint.

A third test has been carried out with four mafic clasts in a 2 m thick rhyolite flow and associated red mudstone overlying the site BA6 amygdaloidal basalt flow; this test may be considered to be an intraformational "pseudo-conglomerate" test, with the clasts having been drawn from the underlying basalt flow. The distribution of these clast mean high-temperature hematite-borne component (Fig. 8*d*) directions appears to be random between clasts and consistent within clasts (Fig. 9*c*), although four clasts are too few to carry out a statistically significant test. However, the $r = 0.302$ is rather small even compared with $r_c = 0.7$ for $n = 5$ (Mardia 1972), so we consider this test as tentatively positive.

Contact test

A 10 m thick basalt flow at site BA5 is underlain by finely laminated purple siltstone, which we sampled at 15 cm, 2 m, and 4 m stratigraphic distances below the flow. Figure 9*d* shows the ChRM directions of the BA5 basalt flow and the siltstones below the contact. The ChRM direction of the baked sediment at 15 cm is similar to that of the overlying basalt flow. The direction of the sample taken 2 m below the flow is only slightly different and possibly also baked by the flow. The sample taken 4 m below the flow is magnetized inversely, so the contact test is positive if we assume that geomagnetic reversal occurred after the deposition of siltstone, but before the cooling of the lava flow. However, we cannot exclude an alternative possibility that the reversal occurred during the deposition of the sediment located from 4 to 2 m from the contact.

Bull Arm Formation, Argentia locality

In the Argentia area, we collected samples from two limbs of an open north–south-hinging anticline (Fig. 3). We have included eight unpublished site mean results from Jordan

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Table 2. Bull Arm Formation mafic rocks, high-temperature component.

Site	Location		DipD (°)	Dip (°)	*N*/*n*	Dg(°)	Ig(°)	*k*	α_{95} (°)	Ds (°)	Is (°)	*k*	α_{95} (°)	VGP		dp (°)	dm (°)
	Latitude (N)	Longitude (W)												(°N)	(°E)		
Bonavista locality																	
BA1	48°27.07′	53°32.48′	92	66	8/8	207.3	–13.9	15.3	14.6	205.6	16.5	15.3	14.6	28.8	97.3	7.8	15.1
BA2*	48°27.07′	53°32.48′	92	66	6/5	99.2	–3.7	10.9	24.3	111.8	–68.5	10.9	24.3	—	—	—	—
BA3	48°27.07′	53°32.48′	92	66	7/6	237.5	9.2	23.5	14.1	194.7	54.2	23.5	14.1	5.8	114.4	13.9	19.8
BA4	48°27.07′	53°32.48′	94	76	6/5	211.9	–16.8	36.1	12.5	208.8	21.6	36.1	12.5	25.1	95.0	7.2	13.6
BA5	48°27.07′	53°32.48′	94	76	6/6	182.1	–24.6	82.9	7.4	207.6	–6.0	82.9	7.4	38.8	90.1	3.7	7.4
BA6	48°23.65′	53°33.33′	79–80	45–60	6/3	52.6	–13.0	18.9	29.2	50.3	–28.4	81.8	13.7	12.4	76.9	8.2	15.0
BA8	48°27.52′	53°31.68′	94	76	8/5	59.2	17.5	27.9	14.8	54.0	–42.8	27.9	14.8	2.2	79.2	11.3	18.3
BA10	48°27.77′	53°31.67′	84	64	6/3	52.0	–8.3	76.5	14.2	18.4	–54.9	76.5	14.2	4.5	111.5	14.3	20.1
BA11	48°27.77′	53°31.67′	84	64	6/3	62.9	–15.6	111.9	11.7	17.9	–67.7	111.9	11.7	–10.3	115.0	16.3	19.6
BA12	48°27.77′	53°31.67′	84	64	6/5	209.3	–10.5	507.2	3.4	201.2	25.5	507.2	3.4	25.3	103.6	2.0	3.7
nine sites				59/44	222.1	–4.4	10.8	16.4	209.8	34.7	10.4	16.8	15.1	98.4	A_{95} = 13.7°		
Argentia locality																	
	West Limb																
BA20	47°16.15′	53°58.79′	254	33	6/6	187.9	62.8	14.9	18.0	229.7	37.7	14.9	18.0	8.3	80.1	12.5	21.2
SJ2	47°16.15′	53°58.79′	300	43	5/5	169.3	62.9	54.0	10.5	258.4	58.6	54.0	10.5	–21.1	71.7	11.6	15.6
SJ3	47°16.15′	53°58.79′	280	43	4/4	15.8	–64.9	95.1	9.5	66.9	–39.3	95.1	9.5	–1.8	67.6	6.8	11.4
	East Limb																
SJ8	47°16.15′	53°55.75′	135	29	8/8	87.3	–44.7	55.8	7.5	53.4	–57.9	55.8	7.5	–8.1	86.7	8.1	11.0
SJ5	47°16.15′	53°55.75′	127	27	5/5	57.7	–24.7	30.8	14.0	43.5	–31.2	30.8	14.0	15.0	83.1	8.8	15.7
SJ6	47°16.15′	53°55.75′	127	27	5/5	66.1	–18.5	74.8	8.9	54.9	–29.5	74.8	8.9	10.1	73.0	5.4	9.8
BA22	47°16.15′	53°55.68′	123	24	8/8	65.8	–20.6	80.1	6.2	55.2	–31.8	80.1	6.2	8.7	73.5	3.9	7.0
seven sites				41/41	231.7	46.9	6.9	24.8	236.2	41.2	30.1	11.2	1.7	76.5	A_{95} = 10.9°		

Note: See footnotes to Table 1. VGP, virtual geomagnetic pole (calculated after tilt corrections); dp, dm, the semi-axes of the cone of confidence about the pole at the 95% probability level.
*Excluded from final calculations (see text).

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Fig. 6. Bull Arm Formation, Bonavista. (*a*) Stereoplot of the site mean directions of ChRM; (*b*) NE upward vectors inverted. Projection and symbols as in Fig. 5.

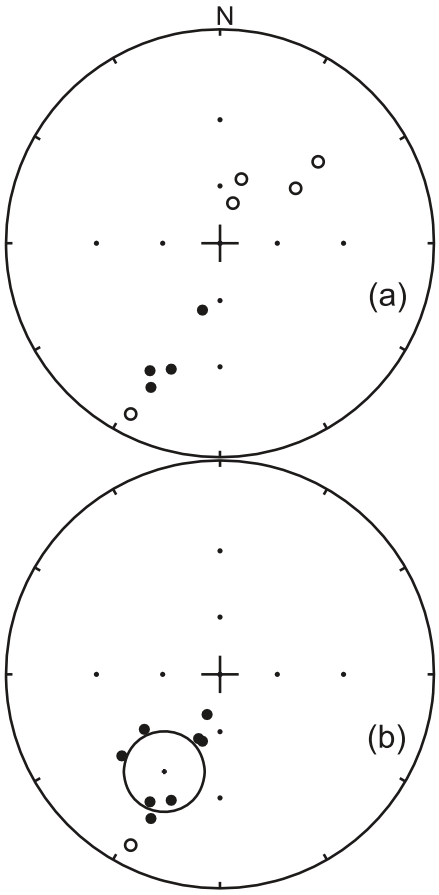
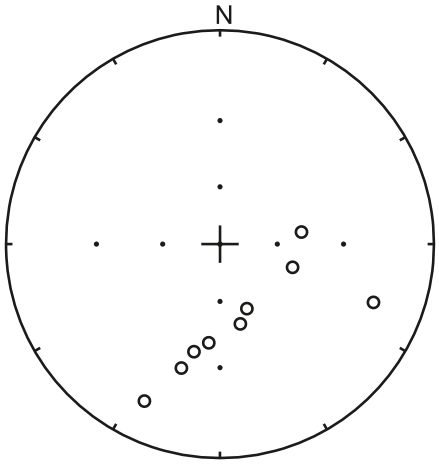


Fig. 7. Bull Arm Formation, site BA7 — elongated distribution, excluded from analyses. Projection and symbols as in Fig. 5.



(1999). Sister undemagnetised specimens from the latter study were analysed in this work and found to provide similar demagnetisation results, so we combined all specimen demagnetisations to augment the original Jordan (1999) pilot study.

Specimens from the Bull Arm Formation in the Argentia locality are sampled from basalt flows and siltstones. As in the Bonavista area, two remanence components were isolated, but in this case the mid-temperature component yields a more consistent direction that is directed northwards and down (Fig. 10) showing a negative fold test (maximum grouping at –20% unfolding). Hence the age of this component is probably post-Silurian. The high-temperature component is bipolar (Table 2) and pre-folding (Fig. 11, maximum grouping at 95% unfolding), with a tilt-corrected mean direction from seven sites of $D = 236.2°$, $I = 41.2°$ ($\alpha_{95} = 11.2°$), including five "inverted" sites.

Crown Hill Formation, Bonavista peninsula

We collected red siltstone and mudstone samples from the upper part of the Crown Hill Formation in a coastal outcrop east of Duntara and in three outcrops along the Duntara – King's Cove road, which represent stratigraphically lower levels (Fig. 2). All strata dip gently to the north-northeast. Thermal demagnetisation of the Crown Hill red beds show dominance of hematite (Fig. 12). The bipolar ChRM is quite stable and consistent (Fig. 13; Table 3). The reversal test of McFadden and McElhinny (1990) is positive (Rc, $\Delta = 3.5°$, $\Delta_c = 15.3°$). Importantly, the stratigraphically higher sites (C15a, C15b, C17) and the topmost sample of the stratigraphically lower site C18 have an exclusively "normal" (northwest, downward) polarity. The samples in the lower part of the section (the rest of site C18 and sites C19a and C19b) have "inverse" (southeast, upward) remanence directions, except for one sample in site C19a. This magnetostratigraphic pattern, supported by the positive reversal test, and the positive regional conglomerate test sampled in the Crown Hill Formation at Red Cliff (Fig. 9*b*), suggests that the ChRM carried by the Crown Hill Formation is primary. The tilt-corrected mean direction from six sites in the Crown Hill red beds is $D = 333.8°$, $I = 42.4°$ ($\alpha_{95} = 14.6°$), including two "inverted" sites (Table 3).

Discussion

Mid-Ediacaran Bull Arm Formation

In this study, we have found that mafic volcanic flows and associated siltstones of the Bull Arm Formation carry a primary hematite-borne ChRM, which is indicated by a positive (Late Silurian – Early Devonian) fold test for the Argentia locality and by three positive conglomerate tests in the Bonavista locality. The Bull Arm Formation at both localities features reversals that are stratigraphically controlled. We consider it likely that the ChRM was acquired soon after the extrusion of these sub-aerial lavas, probably during oxidation associated with their cooling.

The Bull Arm result obtained from the Bonavista locality is from flows that are correlated with Bull Arm volcanic rocks dated at 570 $^{+5}_{-3}$ Ma (U–Pb zircon; O'Brien et al.1989), so we ascribe that age to the Bonavista result (Ta-Table 4). The Bull Arm result from the Argentia locality has a more uncertain age, being only constrained to be pre-folding in age, along with its less clear stratigraphic position with respect to the dated Bull Arm volcanic rocks. The mean remanence directions for the Argentia and Bonavista localities are similar and their circles of confidence overlap (Fig. 14),

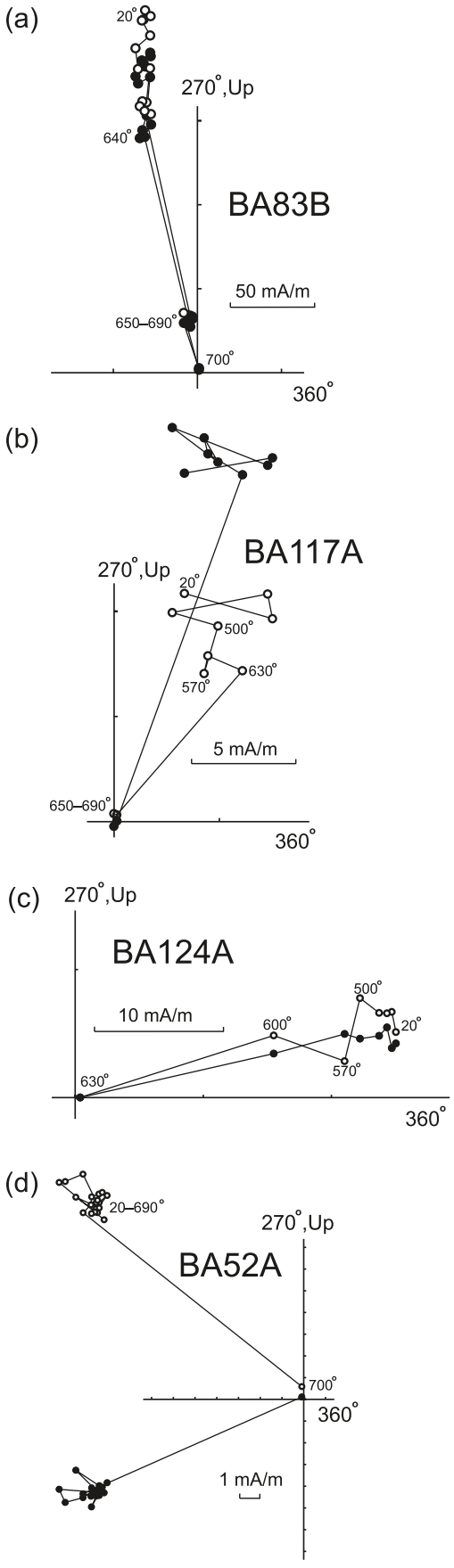


Fig. 8. Bull Arm Formation, Bonavista. Conglomerate test — orthogonal plots. (*a*) Site BA9, hematite-bearing ChRM; (*b*) site BA13, hematite-bearing ChRM; (*c*) site BA13, possibly partly magnetite-bearing ChRM; (*d*) site BA6, hematite-bearing component.

so we consider them to be of similar mid-Ediacaran age. It remains possible that the Argentia locality could be stratigraphically older than the Bonavista locality, but no older than 610 Ma (Table 4), based on regional stratigraphic relationships (Jenness 1963; Dec et al. 1992; O'Brien 1993; Fig. 1).

Late Ediacaran Crown Hill Formation

The result from the ca. 550 Ma Crown Hill Formation indicates that West Avalonia lay at 24.2°S ± 5.3°S at the close of the Ediacaran (Table 4). Notably, the mean direction from the Crown Hill Formation is rotated over 90° clockwise from the mean directions of both Bonavista and Argentia ca. 570 Ma Bull Arm results, despite the overall similarity of the NNE–SSW structural trends in Bonavista and western Avalon peninsulas (Fig. 1), implying tectonic rotation of the whole terrane, or possibly a 570 to 550 Ma large true polar wander (TPW) event (McCausland et al. 2007), with Avalonia lying near the TPW axis of rotation. The Crown Hill mean direction is also, however, strikingly dissimilar to any from the subsequent Paleozoic records in Avalonia (Hodych and Buchan 1998), except for the ca. 442 Ma Midland Minor intrusive suite of East Avalonia, which is similarly anomalous and interpreted to possibly represent a rotated block (Vizan et al. 2003).

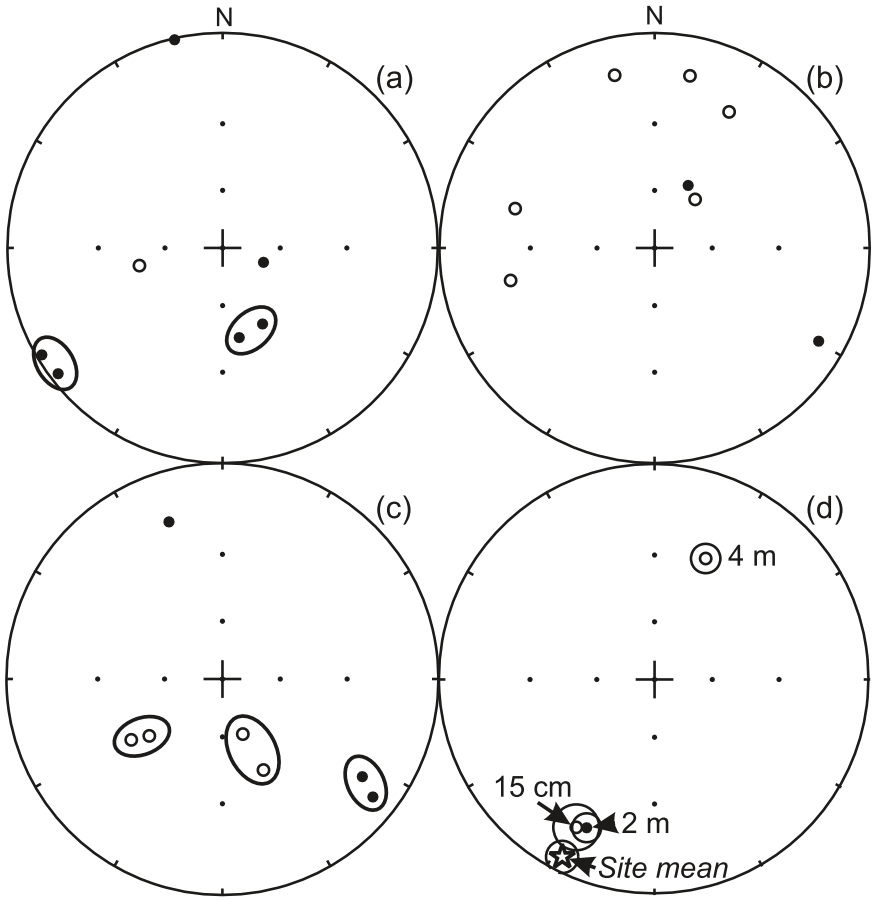
Paleogeographic implications for Avalonia in the Ediacaran

Our new data from Bull Arm and Crown Hill Formations indicate low to medium paleolatitudes for West Avalonia at ~570 and ~550 Ma. Based on the results from this study alone, West Avalonia appears to have resided at low to medium paleolatitudes during the latter part of the Ediacaran, placing the ca. 565 Ma Ediacaran fauna-bearing strata at Mistaken Point and on the Bonavista peninsula (Narbonne and Gehling 2003; O'Brien et al. 2006) at <32°S during their deposition.

We reviewed other published 620–540 Ma paleomagnetic results from both East and West Avalonian terranes in view of more recent geochronological research on rocks that had been studied paleomagnetically (Tucker and Pharaoh 1991; Compston et al. 2002; Schofield et al. 2010). After filtering out Phanerozoic overprints and data without proper demagnetization schemes there are six (including three new results of this study) West Avalonian and five East Avalonian paleomagnetic results (Table 4; Fig. 15). Because of the poor Cambrian paleomagnetic database, the sign of paleomagnetic polarity for Precambrian Avalonia results is unclear, but we will assume a southern hemisphere polarity, as suggested by most published Ediacaran paleogeographic reconstructions (e.g., Dalziel 1997; Cordani et al. 2003; Meert and Torsvik 2003; Murphy et al. 2004; Tohver et al. 2006; McCausland et al. 2007; Li et al. 2008; Pisarevsky et al. 2008).

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Fig. 9. Bull Arm Formation, Bonavista — field tests. (*a*) Conglomerate test in the boulder of the Rocky Harbour Formation conglomerate with pebbles of the Bull Arm volcanics (site BA9), circled directions represent two cylinders drilled from the same cobble; (*b*) conglomerate test in the Crown Hill conglomerate with pebbles of the Bull Arm volcanics (site BA13); (*c*) conglomerate test in the mafic clasts in the rhyolitic matrix in site BA6, circled directions represent two cylinders drilled from the same clast; (*d*) contact test in site BA5, distance from the contact is indicated in metres and centimetres, and star represents the mean (upward) direction for the site. Projection and symbols as in Fig. 5.



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The paleomagnetic results for West and East Avalonia to first order agree with one another, producing similar paleolatitude estimates where ca. 600 and ca. 570 Ma comparisons can be made (Fig. 15). This comparison supports the idea that West and East Avalonia were together throughout the Ediacaran, as expected from geological considerations (e.g., Cocks and Fortey 1982; McKerrow et al. 1992; Murphy et al. 2004; Nance et al. 2008 and references therein).

The ages are not uniformly distributed, with only one relatively high paleolatitude of 55° ± 8° from the Roxbury Formation, New England (Fang et al. 1986) for the time interval between 593 and 570 Ma. Its imprecise age is constrained by the 593 Ma youngest detrital zircon in the lower part of the formation (Thompson et al. 2007) and by the youngest of a range of zircon dates of the ash bed in the overlying Cambridge Argillite, which is 570 Ma (Thompson and Bowring 2000). The Roxbury result was characterized by Thompson et al. (2010) as a possible overprint, but it is apparently supported by a positive conglomerate test for primary remanence, though without details (Fang et al. 1986). We retain the Roxbury result in Table 4 and in Fig. 15 and leave the judgement to the reader and future studies of the Roxbury Formation. All other Avalonia paleolatitudes in Table 4 are below 40°.

We should also note that three results from Uriconian rocks in East Avalonia with ages between 568 and 564 Ma (entries 8–10 in Table 4) all indicate low paleolatitude, but none of these studies provide convincing evidence for primary or near primary remanence. However, the corresponding paleopoles lie apart from the Phanerozoic European Apparent Polar Wander Path (McElhinny and McFadden 2000), so a younger overprint seems to be unlikely.

The paleolatitude drift history of Avalonia can be interpreted in two ways: if the solitary higher paleolatitude result from the Roxbury Formation is primary, then Avalonia would appear to have moved from paleolatitudes of ~30°S at ~600 Ma to ~55°S during the deposition of the Roxbury Formation, and then back to ~35°S or lower paleolatitudes by 570 Ma. Interestingly, the postulated residence of Avalonia at >45°S at some time between 593 and 565 Ma corresponds with the occurrence of glaciogenic sedimentation in Avalonia, in the Boston area (ca. 590 Ma; Thompson et al. 2007) and at Gaskiers, Newfoundland (584–582 Ma; Bowring et al. 2003; Hoffman and Li 2009).

If the result from the Roxbury Formation is not primary or if the New England portion of Avalonia was not strictly amalgamated with the rest of Avalonia at that time, then the

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Fig. 10. Bull Arm Formation, Argentia area. Negative fold test for the magnetite bearing component. (*a*) In situ; (*b*) tilt corrected. Projection and symbols as in Fig. 5.

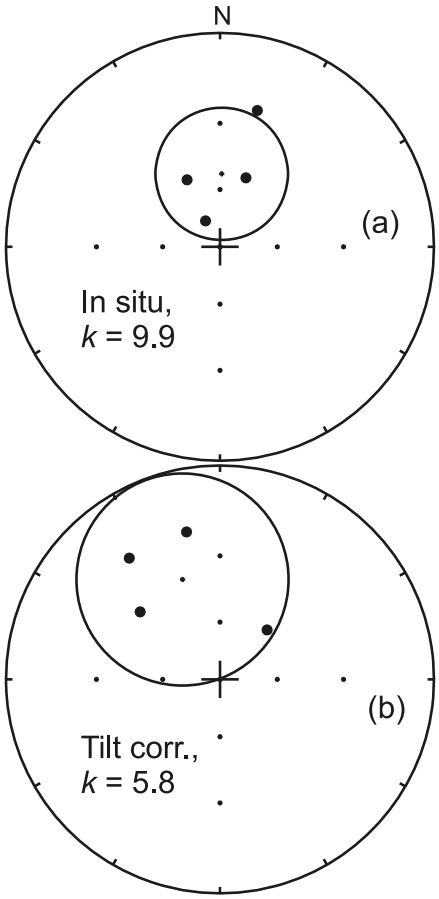
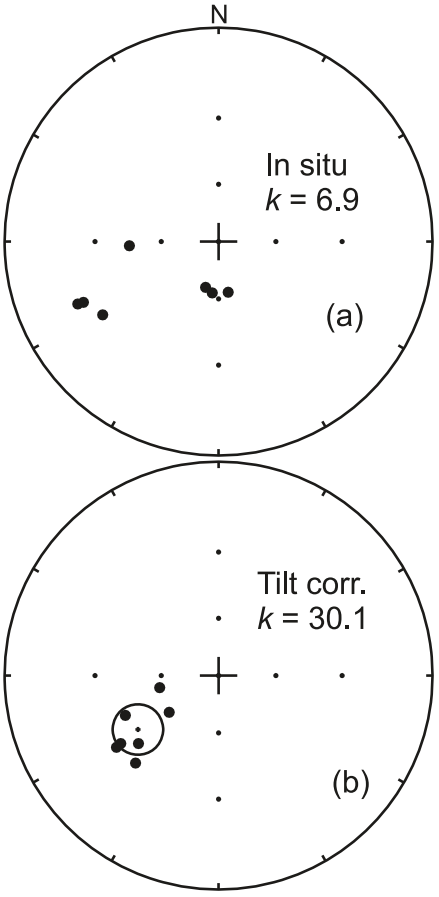


Fig. 11. Bull Arm Formation, Argentia area. Positive fold test for the hematite bearing component. (*a*) In situ; (*b*) tilt corrected. Projection and symbols as in Fig. 5.



remaining paleomagnetic results for Avalonia can be interpreted to represent relatively minor variation in low to mid southerly paleolatitudes from ~600 Ma onwards through the Ediacaran.

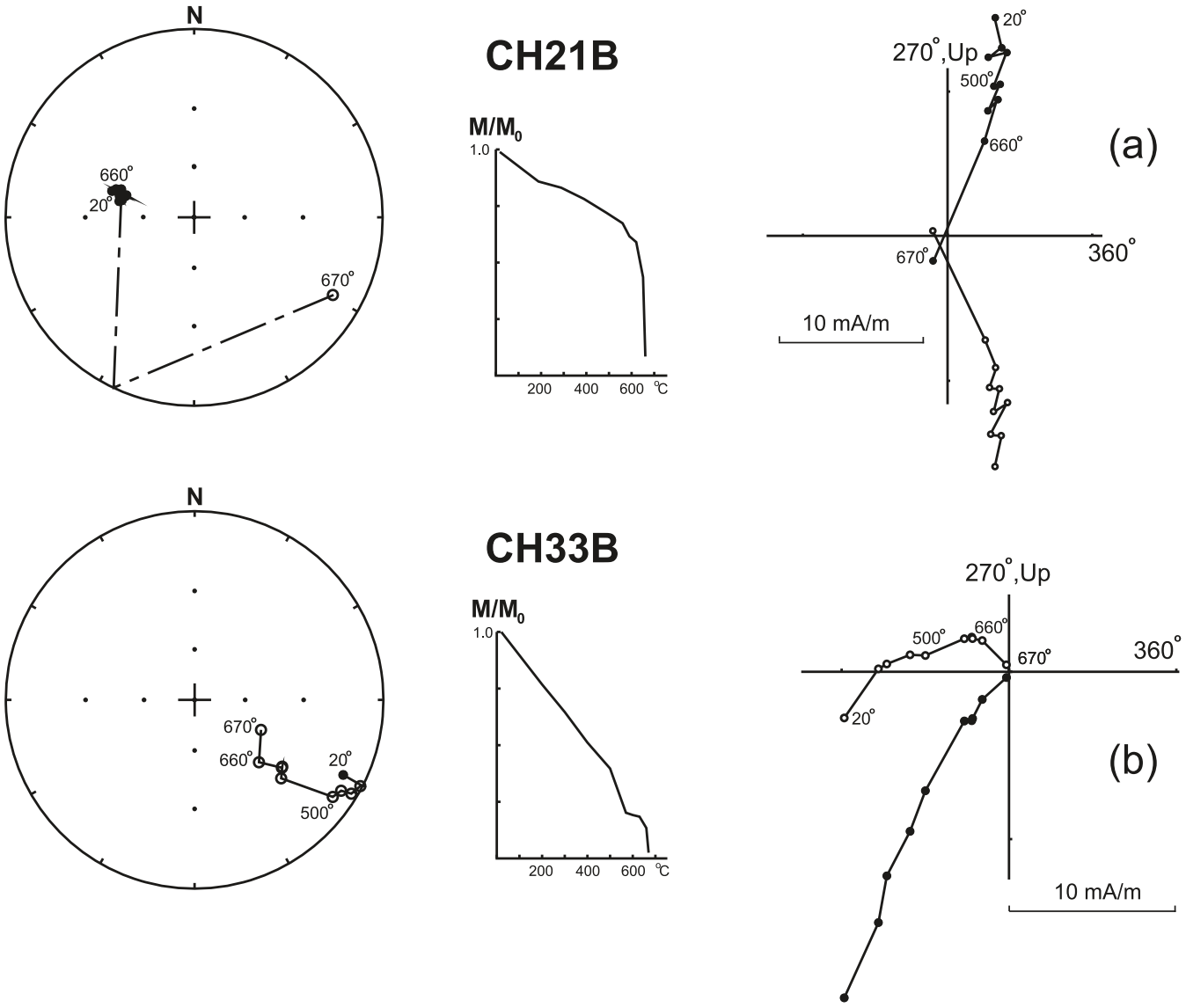
Geodynamic implications from Avalonian paleomagnetic results

Application of the Avalonian paleomagnetic results to Ediacaran global paleogeography depends on the proximity of Avalonia to major continents. Most workers agree that the Avalonian blocks were Peri-Gondwanan terranes and at 600–550 Ma were attached or situated close to the (present) northern margin of Amazonia and (or) West Africa (McNamara et al. 2001; Murphy et al. 2004; Thompson et al. 2007; Nance et al. 2008, 2010 and references therein). These two major continents were probably juxtaposed to each other in a Gondwanan configuration (Trompette 1994, 1997). Unfortunately, Ediacaran paleomagnetic data from Amazonia and West Africa are scarce and controversial, so their paleolatitudes at ~600–550 Ma are unknown (Tohver et al. 2006). In this situation, paleomagnetic data from peri-Gondwanan terranes may provide paleolatitudinal constraints for these West Gondwanan cratons. We shall follow this hypothesis of the West African – Amazonian affinity of Avalonia.

Published global paleogeographic reconstructions for 600–550 Ma vary considerably (e.g., Grunow et al. 1996; Dalziel 1997; Cordani et al. 2003; Pisarevsky et al. 2008 and references therein). The main reason is the Laurentian paleomagnetic database, which contains two sets of equally reliable paleopoles indicating either high- or low-latitude position of Laurentia correspondingly (see a summary of Pisarevsky et al. 2008 and references therein). If all of the Ediacaran paleopoles for Laurentia are primary, an unusually high rate of plate movement or some contribution of TPW is required (McCausland et al. 2007; 2011). Consequently, two alternative sets of reconstructions have been proposed (Murphy et al. 2004; McCausland et al. 2007; 2011; Pisarevsky et al. 2008).

Another possibility involves the dominant contribution from a rapid interchange event in the Earth's moment of inertia tensor (inertial interchange true polar wander (IITPW); Kirschvink et al. 1997). The idea has been further developed (Evans 1998, 2003) and criticized (Meert 1999; Tsai and Stevenson 2007). If the IITPW hypothesis is correct, evidence of similar rapid apparent polar wander (APW) should be present coevally in the rock records of all continents. Schmidt and Williams (2010) demonstrated that there is no large shift in paleolatitude for Australia, although their compilation does show a relatively large APW for that continent during the Ediacaran.

Fig. 12. Thermal demagnetizations of the Crown Hill redbeds. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetisation. See notes for Fig. 3. Projection and symbols as in Fig. 5.



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Recently, Abrajevitch and Van der Voo (2010) suggested that the controversy of Laurentia and Baltica Ediacaran paleopoles could be explained by an alternation of the geomagnetic dipole axis between a geocentric coaxial and an equatorial alignment. This is an interesting idea, but it is in need of further testing. In particular, as for the IITPW, the geocentric equatorial dipole hypothesis requires similar behaviour of paleopoles from other continents.

In general, the low to moderate southerly paleolatitudes of Avalonia through the Ediacaran (Fig. 15) imply that the adjacent Amazonia – West Africa margin of West Gondwana similarly did not experience much change in paleolatitude, thus potentially placing a constraint on global geodynamic models. The result from the Roxbury Formation (Thompson et al. 2007) is the only relatively high paleolatitude for Ediacaran Avalonia (Table 4; Fig. 15) and, consequently, for the "northern" West Gondwana (northern parts of Amazonia and West Africa in present coordinates). Excluding the Roxbury Formation result, the Avalonia paleolatitude distribution supports a "low-latitude" Laurentia model for the global late Neoproterozoic paleogeography (Pisarevsky et al. 2008), because the "high-latitude" Laurentia model requires near-polar position of this area in mid-Ediacaran time, assuming that Laurentia and West Gondwana were closely associated throughout the Ediacaran.

570–550 Ma Avalonia – West Gondwana relations

The mid- and late Ediacaran paleomagnetic results from Avalonia may be compared with results of similar age from West Gondwana, if Amazonia and West Africa are taken to have essentially joined the other elements of West Gondwana by ca. 630 Ma (Trompette 1997; Cordani et al. 2009). We

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Fig. 13. Crown Hill Formation. Stereoplot of the sample mean directions of ChRM. Projection and symbols as in Fig. 5.

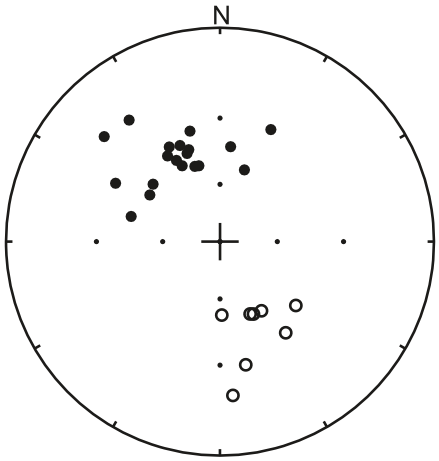


Fig. 14. Bull Arm Formation. Site mean directions in Bonavista (circle) and Argentia (triangle) areas. Projection and symbols as in Fig. 5.

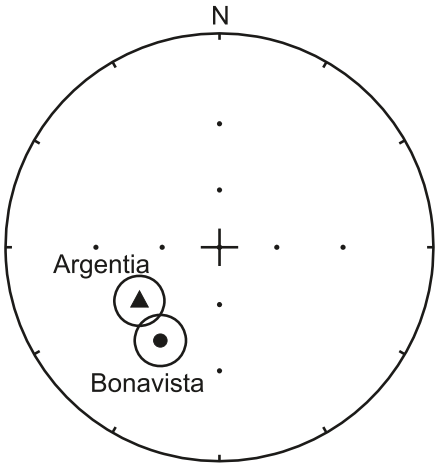


Fig. 15. Ediacaran paleolatitudes of Avalonia.

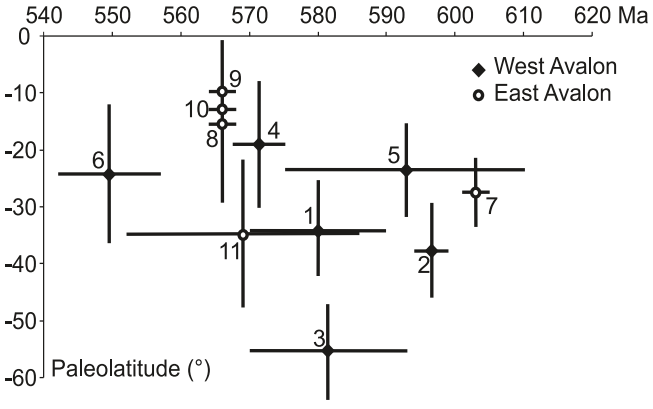


Table 3. Crown Hill Formation red beds, high-temperature (hematite) component.

Site No.	Location		DipD (°)	Dip (°)	*N*/*n*	Dg (°)	Ig (°)	*k*	α_{95} (°)	Ds (°)	Is (°)	*k*	α_{95} (°)	VGP (°N)	VGP (°E)	dp (°)	dm (°)
	Latitude (N)	Longitude (W)															
C15a	48°35.81′	53°21.88′	340–30	12–19	5/4	352.9	59.7	22.6	19.7	355.3	42.3	26.6	18.2	65.6	137.0	13.8	22.4
C15b	48°35.81′	53°21.88′	357–17	17–19	6/5	307.2	43.1	28.7	14.5	318.4	30.9	25.3	15.5	43.5	188.0	9.7	17.3
C17	48°35.43′	53°21.46′	16–44	20–35	6/6	308.7	56.2	180.8	5.0	338.4	44.4	170.6	5.1	61.9	171.2	4.0	6.4
C18	48°35.52′	53°21.67′	28–32	14–21	5/3	282.6	42.9	83.9	13.5	299.8	45.7	76.8	14.2	39.4	214.4	11.5	18.1
C19a*	48°34.56′	53°20.30′	11–31	24–27	4/4	166.2	–59.5	9.9	30.8	178.5	–39.0	11.9	27.7	63.4	129.8	19.7	33.0
C19b	48°34.56′	53°20.30′	11–31	24–27	5/5	126.5	–52.4	33.0	13.5	151.1	–41.4	31.9	13.8	56.4	179.6	10.3	16.8
All samples					33/27	313.9	54.5	16.4	7.1	334.3	41.9	16.8	7.0	58.3	175.5	5.3	8.6
All sites					6/5	306.5	52.4	32.8	13.5	328.6	42.4	24.6	15.7	55.8	183.8	A_{95} = 17.7	

Note: See footnotes to Tables 1and 2. VGP calculated after tilt correction.
*Excluded from the site mean.

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Table 4. Paleolatitudes of West and East Avalonian terranes in the late Neoproterozoic.

#	Rock name, place	Maximum age (Ma)	Minimum age (Ma)	Pallat (°S)	dp (°)	Reference
West Avalonia						
1	Marystown Group, Burin, Newfoundland	590	570	33.6	8.0	McNamara et al. 2001
2	Lynn-Mattapan Volcanics, New England	599	594	37.6	8.4	Thompson et al. 2007
3	Roxbury Formation, New England	593	570	55.1	7.8	Fang et al. 1986; age from Thompson et al. 2007
4	Bull Arm Fm, Bonavista, Newfoundland	575	567	19.1	11.1	This study
5	Bull Arm Fm, Argentia, Newfoundland	610	575	23.6	8.3	This study
6	Crown Hill Fm, Bonavista, Newfoundland	557	542	24.5	11.9	This study
East Avalonia						
7	Caldecote Volcanics, England	605	601	27.4	6.0	Vizan et al. 2003
8	Eastern Uriconian Volc. (north), England	568	564	15.5	13.7	Piper and Strange 1989; age from Tucker and Pharaoh 1991
9	Eastern Uriconian Volc. (south), England	568	564	9.8	5.6	Piper and Strange 1989; age from Tucker and Pharaoh 1991
10	Western Uriconian Volc., England	568	564	12.8	8.8	Piper and Strange 1989; age from Tucker and Pharaoh 1991
11	Post-Uriconian Dolerites, England	586	552	34.7	13.0	Piper 1979; age from Tucker and Pharaoh 1991; Compston et al. 2002

Note: Pallat, paleolatitude, north or south; dp, lesser semi-axis of the confidence oval, giving estimated error in the paleolatitude.

Fig. 16. Possible paleolatitudinal relation of West Avalonia to the Amazonia – West Africa margin of West Gondwana for (*a*) 570 Ma and (*b*) 550 Ma. The shaded region marks the paleolatitude ranges found for the 570 Ma Bull Arm Formation and the 550 Ma Crown Hill Formation. Constituent cratons for West Gondwana are denoted as AM, Amazonia; C-SF, Congo – São Francisco; RP, Rio de la Plata; WAF, West Africa. Parts of South America and its paleopoles have been rotated into West African coordinates using the Euler rotation poles of McElhinny et al. (2003) — for Rio de la Plata: 48.8° N, 324.9°E, 52.8 °CCW. West Gondwana paleopoles are (1) PA, combined 579 Ma pole, Rio de la Plata (16.9°S, 250.9°E, A_{95} = 18.5°; Sanchez-Bettucci and Rapalini, 2002; Oyhantçabal et al. 2007); (2) LB, Los Barrientos, Rio de la Plata (15.1°S, 252.6°E, A_{95} = 12.4°; Rapalini, 2006); (3) SD, Sinyai metadolerite dykes, Congo (20°S, 319°E, A_{95} = 4°; Meert and Van der Voo, 1996).

note, however, that the assembly of West Gondwana by some interpretations did not take place until the Cambrian (Trindade et al. 2006; Tohver et al. 2010).

The mid-Ediacaran, paleomagnetic database for other West Gondwanan cratons is also poor. Most results are poorly dated and their primary remanence is not proven. Moreover, some of these results are from metamorphic belts, and it thus is not clear if the corresponding terranes already were accreted on the cratonic margins. However, the SA2 pole in Rio de la Plata craton (Sanchez-Bettucci and Rapalini 2002; 579 Ma Ar/Ar cooling age from Oyhantçabal et al. 2007) probably can be used for at least paleolatitudinal constraint. The expected paleolatitude position of West Avalonia as provided by the Bull Arm Formation result (Table 4) likely lies

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within the shaded region of Fig. 16*a*, between 8°S and 32°S along the join between the Amazonian and West African margins of West Gondwana.

In the late Ediacaran (Fig. 16*b*), the ca. 550 (± 50) Ma Los Barrientos siltstones (Rapalini 2006) and the 547 Ma Sinyai metadolerite dykes (Meert and Van der Voo 1996) are taken to represent West Gondwana. The coeval Crown Hill Formation paleolatitude is plotted as the shaded region between 18°S and 30°S along the West African margin of Gondwana.

The shift in West Avalonia's cratonic affinity along the West Africa stems mainly from the apparent southwards shift of West Gondwana between 570 and 550 Ma. The relative motion between Avalonia and the Amazonia – West Africa margin is represented schematically in Fig. 16*b* as a dextral transcurrent motion of about 1000 km along the margin. These results are broadly consistent with models in which Avalonian magmatism is terminated between 600 and 540 Ma by the generation of a San Andreas-style transform fault (Murphy et al. 2000) and with detrital zircon data from Ediacaran aged clastic rocks which indicate a shift from Amazonian to West African provenance in the Ediacaran (Gutiérrez-Alonso et al. 2005).

Conclusions

1. Paleomagnetic study of the ~570 Ma Bull Arm volcanics in Bonavista and Argentia localities of the Avalon Zone, Newfoundland reveal a consistent bipolar ChRM, carried by hematite, that is likely primary, as shown by several positive field tests and by apparent magnetostratigraphic pattern in the studied section. Paleomagnetic directions are slightly different in Bonavista and Argentia, which may because of to slight structural differences between the two localities, or possibly age difference between the studied rocks. A less stable magnetite-borne component in the Argentia locality postdates Silurian folding.
2. The paleomagnetic study of the ~550 Ma Crown Hill red beds in the northern part of the Bonavista peninsula reveal a stable bipolar hematite borne ChRM supported by positive reversal test and by a magnetostratigraphic pattern in the studied section.
3. These results suggest a low-latitude position of Avalonia in the mid- to late Ediacaran, during the period when complex Ediacaran fauna became established in West Avalonia.
4. Ediacaran paleolatitudes of Avalonia indicate only moderate-scale drift of this terrane and suggest that the Amazonia – West Africa portion of Gondwana underwent little drift during this time, assuming Avalonia maintained a relatively fixed relationship with those cratons.
5. Alternatively, a ~1000 km dextral motion of Avalonia with respect to Amazonia – West Africa can be inferred from the comparison of Avalonia paleolatitudes with the position of West Gondwana at ~570 and ~550 Ma, assuming West Gondwana was assembled prior to 570 Ma.

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References

Abrajevitch, A., and Van der Voo, R. 2010. Incompatible Ediacaran paleomagnetic directions suggest an equatorial geomagnetic dipole hypothesis. Earth and Planetary Science Letters, **293**(1–2): 164–170. doi:10.1016/j.epsl.2010.02.038.

Abrajevitch, A., Van der Voo, R., Bazhenov, M.L., Levashova, N.M., and McCausland, P.J.A. 2008. The role of the Kazakhstan orocline in the late Paleozoic amalgamation of Eurasia. Tectonophysics, **455**(1–4): 61–76. doi:10.1016/j.tecto.2008.05.006.

Bowring, S.A., Myrow, P., Landing, E., Remezani, J., and Grotzinger, J. 2003. Geochronological constraints on terminal Neoproterozoic events and the rise of metazoans. Geophysical Research Abstracts, **5**: Abstract 13219.

Cawood, P.A., and Pisarevsky, S.A. 2006. Was Baltica right way up or upside down in the Neoproterozoic? Journal of the Geological Society, **163**(5): 753–759. doi:10.1144/0016-76492005-126.

Cocks, L.R.M., and Fortey, R.A. 1982. Faunal evidence for oceanic separations in the Palaeozoic of Britain. Journal of the Geological Society, **139**(4): 465–478. doi:10.1144/gsjgs.139.4.0465.

Collins, A.S., and Pisarevsky, S.A. 2005. Amalgamating Eastern Gondwana: the evolution of the Circum-Indian orogens. Earth-Science Reviews, **71**(3-4): 229–270. doi:10.1016/j.earscirev.2005.02.004.

Compston, W., Wright, A.E., and Toghill, P. 2002. Dating the Late Precambrian volcanicity of England and Wales. Journal of the Geological Society, **159**(3): 323–339. doi:10.1144/0016-764901-010.

Coney, P.J., Jones, D.L., and Monger, J.W.H. 1980. Cordilleran suspect terranes. Nature, **288**(5789): 329–333. doi:10.1038/288329a0.

Cordani, U.G., D'Agrella-Filho, M.S., Brito-Neves, B.B., and Trindade, R.I.F. 2003. Tearing up Rodinia: the Neoproterozoic palaeogeography of South American cratonic fragments. Terra Nova, **15**(5): 350–359. doi:10.1046/j.1365-3121.2003.00506.x.

Cordani, U.G., Teixeira, W., D'Agrella-Filho, M.S., and Trindade, R.I.F. 2009. The position of the Amazonian Craton in supercontinents. Gondwana Research, **15**(3–4): 396–407. doi:10.1016/j.gr.2008.12.005.

Dallmeyer, R.D., Hussey, E.M., O'Brien, S.J., and O'Driscoll, C.F. 1983. Chronology of tectonothermal activity in the western Avalon Zone of the Newfoundland Appalachians. Canadian Journal of Earth Sciences, **20**(3): 355–363. doi:10.1139/e83-034.

Dalziel, I.W.D. 1997. Neoproterozoic–Paleozoic geography and tectonics: Review, hypothesis, environmental speculation. Geological Society of America Bulletin, **109**(1): 16–42. doi:10.1130/0016-7606(1997)109<0016:ONPGAT>2.3.CO;2.

Dec, T., O'Brien, S.J., and Knight, I. 1992. Late Precambrian volcaniclastic deposits of the Avalonian Eastport Basin (New-

Can. J. Earth Sci. Downloaded from www.nrcresearchpress.com by University of Western Australia on 05/28/12
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foundland Appalachians): Petrofacies, detrital clinopyroxene geochemistry and paleotectonic implications. Precambrian Research, **59**(3–4): 243–262. doi:10.1016/0301-9268(92)90059-W.

Dunning, G.R., O'Brien, S.J., Colman-Sadd, S.P., Blackwood, R.F., Dickson, W.L., O'Neill, P.P., and Krogh, T.E. 1990. Silurian Orogeny in the Newfoundland Appalachians. The Journal of Geology, **98**(6): 895–913. doi:10.1086/629460.

Evans, D.A. 1998. True polar wander, a supercontinental legacy. Earth and Planetary Science Letters, **157**(1–2): 1–8. doi:10.1016/S0012-821X(98)00031-4.

Evans, D.A.D. 2003. True polar wander and supercontinents. Tectonophysics, **362**(1–4): 303–320. doi:10.1016/S0040-1951(02)000642-X.

Fang, W., Van der Voo, R., and Johnson, R.J.E. 1986. Eocambrian paleomagnetism of the Boston basin: evidence for a displaced terrane. Geophysical Research Letters, **13**(13): 1450–1453. doi:10.1029/GL013i013p01450.

Fisher, R.A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences, **217**(1130): 295–305. doi:10.1098/rspa.1953.0064.

Friedl, G., Finger, F., McNaughton, N.J., and Fletcher, I.R. 2000. Deducing the ancestry of terranes: SHRIMP evidence for South America-derived Gondwana fragments in central Europe. Geology, **28**(11): 1035–1038. doi:10.1130/0091-7613(2000)28<1035:DTAOTS>2.0.CO;2.

Graham, J.W. 1949. The stability and significance of magnetism in sedimentary rocks. Journal of Geophysical Research, **54**(2): 131–167. doi:10.1029/JZ054i002p00131.

Grunow, A., Hanson, R., and Wilson, T. 1996. Were aspects of Pan-African deformation linked to Iapetus opening? Geology, **24**(12): 1063–1066. doi:10.1130/0091-7613(1996)024<1063:WAOPAD>2.3.CO;2.

Gutiérrez-Alonso, G., Fernández-Suárez, J., Collins, A.S., Abad, I., and Nieto, F. 2005. Amazonian Mesoproterozoic basement in the core of the Ibero–Armorican Arc: $^{40}Ar/^{39}Ar$ detrital mica ages complement the zircon's tale. Geology, **33**(8): 637–640. doi:10.1130/G21485.1.

Hodych, J.P., and Buchan, K.L. 1998. Paleomagnetism of the ca. 440 Ma Cape St Mary's sills of the Avalon Peninsula of Newfoundland: implications for Iapetus Ocean closure. Geophysical Journal International, **135**(1): 155–164.

Hoffman, P.F., and Li, Z.-X. 2009. A palaeogeographic context for Neoproterozoic glaciation. Palaeogeography, Palaeoclimatology, Palaeoecology, **277**(3–4): 158–172. doi:10.1016/j.palaeo.2009.03.013.

Holdsworth, R.E. 1991. The geology and structure of the Gander-Avalon boundary zone in northeastern Newfoundland. *In* Current research. Newfoundland Department of Mines and Energy, Report 91-1. pp 109–126.

Jenness, S.E. 1963. Terra Nova and Bonavista map areas, Newfoundland. Geological Survey of Canada, Memoir 327. 184 p.

Jordan, S. 1999. A reconnaissance paleomagnetic study of the Bull Arm Formation of the Western Avalon peninsula. Unpublished B. Sc. thesis, Memorial University, St. John's, NL. 103 p.

King, A.F. 1988. Geology of the Avalon Peninsula, Newfoundland (parts of 1K, 1L, 1M, 1N and 2C). Newfoundland Department of Mines, Mineral Development Division, Map 88-01, scale 1 : 250 000.

Kirschvink, J.L. 1980. The least squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Kirschvink, J.L., Ripperdan, R.L., and Evans, D.A. 1997. Evidence for large-scale reorganization of Early Cambrian continental masses by inertial interchange true polar wander. Science, **277**(5325): 541–545. doi:10.1126/science.277.5325.541.

Knoll, A.H., Walter, M.R., Narbonne, G.M., and Christie-Blick, N. 2004. A new period for the Geologic Time Scale. Science, **305**(5684): 621–622. doi:10.1126/science.1098803. PMID:15286353.

Kroner, A., and Hegner, E. 1998. Geochemistry, single zircon ages and Sm–Nd systematisc of granitoid rocks from the Gory Sowie (Owl Mts), Polish West Sudetes: evidence for early Paleozoic arc related plutonism. Journal of the Geological Society, **155**(4): 711–724. doi:10.1144/gsjgs.155.4.0711.

Landing, E. 2005. Early Paleozoic Avalon–Gondwana unity: An obituary-response to "Palaeontological evidence bearing on global Ordovician–Silurian continental reconstructions" by R.A. Fortey and L R.M. Cocks. Earth-Science Reviews, **69**(1–2): 169–175. doi:10.1016/j.earscirev.2004.10.002.

Li, Z.X., Bogdanova, S.V., Collins, A., Davidson, A., De Waele, B., Ernst, R.E., et al. 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Research, **160**(1–2): 179–210. doi:10.1016/j.precamres.2007.04.021.

Mardia, K.V. 1972. Statistics of directional data. Academic Press, London, England. 357 p.

McCausland, P.J.A., and Hodych, J.P. 1998. Paleomagnetism of the 550 Ma Skinner Cove volcanics of western Newfoundland and the opening of the Iapetus Ocean. Earth and Planetary Science Letters, **163**(1-4): 15–29. . doi:10.1016/S0012-821X(98)00171-X.

McCausland, P.J.A., Van der Voo, R., and Hall, C.M. 2007. Circum-Iapetus paleogeography of the Precambrian-Cambrian transition with a new paleomagnetic constraint from Laurentia. Precambrian Research, **156**(3–4): 125–152. doi:10.1016/j.precamres.2007.03.004.

McCausland, P.J.A., Hankard, F., Van der Voo, R., and Hall, C.M. 2011. Ediacaran paleogeography of Laurentia: Paleomagnetism and $^{40}Ar–^{39}Ar$ geochronology of the 583 Ma Baie des Moutons syenite, Quebec. Precambrian Research, **187**(1–2): 58–78. doi:10.1016/j.precamres.2011.02.004.

McElhinny, M.W., and Lock, J. 1996. IAGA paleomagnetic databases with Access. Surveys in Geophysics, **17**(5): 575–591. doi:10.1007/BF01888979.

McElhinny, M.W., and McFadden, P.L. 2000. Paleomagnetism: continents and oceans. Academic Press, San Diego, Calif. 386 p.

McElhinny, M.W., Powell, C.McA., and Pisarevsky, S.A. 2003. Paleozoic terranes of Eastern Australia and the drift history of Gondwana. Tectonophysics, **362**(1-4): 41–65. doi:10.1016/S0040-1951(02)00630-3.

McFadden, P.L., and McElhinny, M.W. 1990. Classification of the reversal test in palaeomagnetism. Geophysical Journal International, **103**(3): 725–729. doi:10.1111/j.1365-246X.1990.tb05683.x.

McKerrow, W.S., Scotese, C.F., and Brasier, M.D. 1992. Early Cambrian continental reconstructions. Journal of the Geological Society, **149**(4): 599–606. doi:10.1144/gsjgs.149.4.0599.

McNamara, A.K., Mac Niocaill, C., van der Pluijm, B.A., and Van der Voo, R. 2001. West African proximity of the Avalon terrane in the latest Precambrian. Geological Society of America Bulletin, **113**(9): 1161–1170. doi:10.1130/0016-7606(2001)113<1161:WAPOTA>2.0.CO;2.

Meert, J.G. 1999. A paleomagnetic analysis of Cambrian true polar wander. Earth and Planetary Science Letters, **168**(1–2): 131–144. doi:10.1016/S0012-821X(99)00042-4.

Meert, J.G., and Torsvik, T.H. 2003. The making and unmaking of a supercontinent: Rodinia revisted. Tectonophysics, **375**(1–4): 261–288. doi:10.1016/S0040-1951(03)00342-1.

Meert, J.G., and Van der Voo, R. 1996. Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai dolerite, Kenya: implications for Gondwana assembly. The Journal of Geology, **104**(2): 131–142. doi:10.1086/629810.

Can. J. Earth Sci. Downloaded from www.nrcresearchpress.com by University of Western Australia on 05/28/12
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Meert, J.G., and Van der Voo, R. 2001. Comment on "New palaeomagnetic result from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian" by S.A. Pisarevsky, R.A. Komissarova and A.N. Khramov. Geophysical Journal International, **146**(3): 867–870. doi:10.1046/j.0956-540x.2001.01474.x.

Murphy, J.B., and Nance, R.D. 1989. A model for the evolution of the Avalonian–Cadomian belt. Geology, **17**: 735–738. doi:10.1130/0091-7613(1989)017<0735:MFTEOT>2.3.CO;2.

Murphy, J.B., and Nance, R.D. 2002. Nd–Sm isotopic systematics as tectonic tracers: an example from West Avalonia, Canadian Appalachians. Earth-Science Reviews, **59**(1–4): 77–100. doi:10.1016/S0012-8252(02)00070-3.

Murphy, J.B., Strachan, R.A., Nance, R.D., Parker, K.D., and Fowler, M.B. 2000. Proto-Avalonia: A 1.2 to 1.0 Ga tectonothermal event and constraints for the evolution of Rodinia. Geology, **28**(12): 1071–1074. doi:10.1130/0091-7613(2000)28<1071:PAGTEA>2.0.CO;2.

Murphy, J.B., Pisarevsky, S.A., Nance, R.D., and Keppie, J.D. 2004. Neoproterozoic – early Paleozoic evolution of peri-Gondwanan terranes: implications for Laurentia-Gondwana connections. International Journal of Earth Sciences, **93**(5): 659–682. doi:10.1007/s00531-004-0412-9.

Nairn, A.E.M., Frost, C.V., and Light, B.G. 1959. Palaeomagnetism of certain rocks from Newfoundland. Nature, **183**(4661): 596–597. doi:10.1038/183596a0.

Nance, R.D., Murphy, J.B., Strachan, R.A., Keppie, J.D., Gutiérrez-Alonso, G., Fernández-Suárez, J., et al. 2008. Neoproterozoic-early Paleozoic paleogeography of the peri-Gondwanan terranes: Amazonian versus West African connections. *In* The boundaries of the West African craton. *Edited by* E. Nasser and J-P. Liégeois. Geological Society (of London), Special Publication 297. pp. 245–383.

Nance, R.D., Gutiérrez-Alonso, G., Keppie, J.D., Linnemann, U., Murphy, J.B., Quesada, C., Strachan, R.A., and Woodcock, N. 2010. Evolution of the Rheic Ocean. Gondwana Research, **17**(2–3): 194–222. doi:10.1016/j.gr.2009.08.001.

Narbonne, G.M., and Gehling, J.G. 2003. Life after snowball: the oldest complex Ediacaran fossils. Geology, **31**(1): 27–30. doi:10.1130/0091-7613(2003)031<0027:LASTOC>2.0.CO;2.

Normore, L.S. 2010. Geology of the Bonavista map area (NTS 2C/11), Newfoundland. *In* Current research (2010). Newfoundland and Labrador Department of Natural Resources, Geological Survey, Report 10-1. pp. 281–301.

O'Brien, S.J. 1993. A preliminary account of geological investigations in the Clode Sound – Goose Bay region, Bonavista Bay, Newfoundland (NTS 2C5/NW and 2D8/NE). *In* Current research. Newfoundland Department of Mines and Energy, Geological Survey Branch, Report 93-1. pp. 293–309.

O'Brien, S.J., and King, A.F. 2002. Neoproterozoic stratigraphy of the Bonavista Peninsula: preliminary results, regional correlations and implications for sediment-hosted stratiform copper exploration in the Newfoundland Avalon Zone. *In* Current research (2002). Newfoundland Department of Mines and Energy, Geological Survey, Report 02-1. pp. 229–244.

O'Brien, S.J., and King, A.F. 2005. Late Neoproterozoic (Ediacaran) stratigraphy of Avalon Zone sedimentary rocks, Bonavista Peninsula, Newfoundland Avalon Zone. *In* Current research (2005) Newfoundland Department of Natural Resources, Geological Survey, Report 05-1. pp. 101–113.

O'Brien, S.J., and Knight, I. 1988. Avalonian geology of southwest Bonavista Bay: Parts of the St. Brendans's (2C/13) and Eastport (2C/12) map areas. *In* Current research. Newfoundland Department of Mines, Mineral Development Division, Report 88-1. pp. 193–205.

O'Brien, S.J., Dunning, G.R., Knight, I., and Dec, T. 1989. Late Precambrian Geology of the North Shore of Bonavista Bay, Clode Sound to Lockers Bay. *In* Report of activities. Geological Survey Branch, Department of Mines and Energy. pp. 49–50.

O'Brien, S.J., Strong, D.F., and King, A.F. 1990. The Avalon Zone type area: southeastern Newfoundland Appalachians. *In* Avalonian and Cadomian geology of the North Atlantic. *Edited by* R.A. Strachan and G.K. Taylor. Blackies and Son, Ltd., Glasgow, Scotland. pp. 166–194.

O'Brien, S.J., O'Driscoll, C.F., Greene, B.A., and Tucker, R.D. 1995. Pre-Carboniferous geology of the Connaigre Peninsula and the adjacent coast of Fortune Bay, southern Newfoundland. *In* Current research. Newfoundland Department of Natural Resources, Geological Survey, Report 95-1. pp. 267–298.

O'Brien, S.J., O'Brien, B.H., Dunning, G.R., and Tucker, R.D. 1996. Late Neoproterozoic Avalonian and related peri-Gondwanan rocks of the Newfoundland Appalachians. *In* Avalonian and related Per-Gondwanan terranes of the Circum-North Atlantic. *Edited by* R.D. Nance and M.D. Thompson. Geological Society of America, Boulder, Colo., Special Paper 304. pp. 9–28.

O'Brien, S.J., King, A.F., and Hofmann, H.J. 2006. Lithostratigraphic and biostratigraphic studies on the Eastern Bonavista peninsula: an update. *In* Current research (2006). Newfoundland and Labrador Department of Natural Resources, Geological Survey, Report 06-1. pp. 257–263.

Oyhantçabal, P., Siegesmund, S., Wemmer, K., Frei, R., and Layer, P. 2007. Post-collisional transition from calc-alkaline to alkaline magmatism during transcurrent deformation in the southernmost Dom Feliciano Belt (Braziliano–Pan-African, Uruguay). Lithos, **98**(1-4): 141–159. doi:10.1016/j.lithos.2007.03.001.

Piper, J.D.A. 1979. Palaeomagnetic study of Late Precambrian rocks of the Midland craton of England and Wales. Physics of the Earth and Planetary Interiors, **19**(1): 59–72. doi:10.1016/0031-9201(79)90090-6.

Piper, J.D.A., and Strange, T.M. 1989. A palaeomagnetic study of the Charnian, Caldecote and Uriconian volcanics and plutons, central England. Geological Journal, **24**(4): 331–357. doi:10.1002/gj.3350240407.

Pisarevsky, S.A. 2005. New edition of the Global Paleomagnetic Database. EOS Transactions of the American Geophysical Union, **86**(17): 170. doi:10.1029/2005EO170004.

Pisarevsky, S.A., Komissarova, R.A., and Khramov, A.N. 2000. New palaeomagnetic results from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian. Geophysical Journal International, **140**(3): 598–610. doi:10.1046/j.1365-246X.2000.t01-1-00056.x.

Pisarevsky, S.A., Komissarova, R.A., and Khramov, A.N. 2001. Reply to the comment of J.G. Meert and R. Van der Voo "New palaeomagnetic results from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian". Geophysical Journal International, **146**(3): 871–873. doi:10.1046/j.0956-540x.2001.01475.x.

Pisarevsky, S.A., Murphy, J.B., Cawood, P.A., and Collins, A.S. 2008. Late Neoproterozoic and Early Cambrian palaeogeography: models and problems. *In* West Gondwana: Pre-Cenozoic correlations across the South Atlantic region. *Edited by* R.J. Pankhurst, R.A.J. Trouw, B.B. Brito Neves, and de M.J. Wit. Geological Society (of London), Special Publications 294. pp. 9–31.

Pollock, J.C., Hibbard, J.P., and Sylvester, P.J. 2009. Early Ordovician rifting of Avalonia and birth of the Rheic Ocean: U–Pb detrital zircon constraints from Newfoundland. Journal of the Geological Society, **166**(3): 501–515. doi:10.1144/0016-76492008-088.

Rapalini, A.E. 2006. New late Proterozoic paleomagnetic pole for the Rio de la Plata craton: implications for Gondwana. Precambrian Research, **147**(3–4): 223–233. doi:10.1016/j.precamres.2006.01.016.

Sanchez-Bettucci, L., and Rapalini, A.E. 2002. Paleomagnetism of the Sierra de las Animas complex, southern Uruguay: its implications in the assembly of Western Gondwana. Precambrian Research, **118**(3–4): 243–265. doi:10.1016/S0301-9268(02)00114-6.

Schmidt, P., and Williams, G.E. 2010. Ediacaran palaeomagnetism and apparent polar wander path for Australia: no large true polar wander. Geophysical Journal International, **182**(2): 598–610. doi:10.1111/j.1365-246X.2010.04652.x.

Schofield, D., and D'Lemos, R.S. 2000. Granite petrogenesis in the Gander Zone, NE Newfoundland: mixing of melts from multiple sources and the role of lithospheric delamination. Canadian Journal of Earth Sciences, **37**(4): 535–547. doi:10.1139/e99-116.

Schofield, D.I., Millar, I.L., Wilby, P.R., and Evans, J.A. 2010. A new, high precision U–Pb date from the oldest known rocks in southern Britain. Geological Magazine, **147**(01): 145–150. doi:10.1017/S001675680999063X.

Thompson, M.D., and Bowring, S.A. 2000. Age of the Squantum "Tillite", Boston Basin, Massachusetts: U–Pb zircon constraints on terminal Neoproterozoic glaciation. American Journal of Science, **300**(8): 630–655. doi:10.2475/ajs.300.8.630.

Thompson, M.D., Grunow, A.M., and Ramezani, J. 2007. Late Neoproterozoic paleogeography of the Southeastern New England Avalon Zone: Insights from U–Pb geochronology and paleomagnetism. Geological Society of America Bulletin, **119**(5–6): 681–696. doi:10.1130/B26014.1.

Thompson, M.D., Grunow, A.M., and Ramezani, J. 2010. Cambro-Ordovician paleogeography of the Southeastern New England Avalon Zone: Implication for Gondwana breakup. Geological Society of America Bulletin, **109**: 16–42. doi:10.113/B26581.1.

Tohver, E., D'Agrella-Filho, M.S., and Trindade, R.I.F. 2006. Paleomagnetic record of Africa and South America for the 1200–500 Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambrian Research, **147**(3–4): 193–222. doi:10.1016/j.precamres.2006.01.015.

Tohver, E., Trindade, R.I.F., Solum, J.G., Hall, C.M., Riccomini, C., and Nogueira, A.C. 2010. Closing the Clymene ocean and bending a Brasiliano belt: evidence for the Cambrian formation of Gondwana, southeast Amazon craton. Geology, **38**(3): 267–270. doi:10.1130/G30510.1.

Trindade, R.I.F., D'Agrella-Filho, M.S., Epof, I., and Brito-Neves, B.B. 2006. Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana. Earth and Planetary Science Letters, **244**(1–2): 361–377. doi:10.1016/j.epsl.2005.12.039.

Trompette, R. 1994. Geology of Western Gondwana (2000–500 Ma). A.A. Balkema, Rotterdam and Brookfield. 350 p.

Trompette, R. 1997. Neoproterozoic (~600 Ma) aggregation of Western Gondwana: a tentative scenario. Precambrian Research, **82**(1–2): 101–112. doi:10.1016/S0301-9268(96)00045-9.

Tsai, V.C., and Stevenson, D.J. 2007. Theoretical constraints on true polar wander. Journal of Geophysical Research, **112**(B5): B05415. doi:10.1029/2005JB003923.

Tucker, R.D., and McKerrow, W.S. 1995. Early Paleozoic chronology: A review in light of new U–Pb zircon ages from Newfoundland and Britain. Canadian Journal of Earth Sciences, **32**(4): 368–379. doi:10.1139/e95-032.

Tucker, R.D., and Pharaoh, T.C. 1991. U–Pb zircon ages for Late Precambrian igneous rocks in southern Britain. Journal of the Geological Society, **148**(3): 435–443. doi:10.1144/gsjgs.148.3.0435.

van Staal, C.R., Dewey, J.F., Mac Niocaill, C., and McKerrow, W.S. 1998. The Cambrian-Silurian tectonic evolution of the Northern Appalachians and British Caledonides: history of a complex, west and southwest Pacific-type segment of Iapetus. *In* The past is the key to the present. *Edited by* D. Blundell and A.C. Scott. Geological Society (of London), Special Publication 143. pp. 199–242.

van Staal, C.R., Whalen, J.B., Valverde-Vaquero, P., Zagorevski, A., and Rogers, N. 2009. Pre-Carboniferous, episodic accretion-related, orogenesis along the Laurentian margin of the northern Appalachians. *In* Ancient orogens and modern analogues. *Edited by* J.B. Murphy, J.D. Keppie, and A.J. Hynes. Geological Society (of London), Special Publication 327. pp. 271–316.

Vizan, H., Carney, J.N., Turner, P., Ixer, R.A., Tomasso, M., Mullen, R.P., and Clarke, P. 2003. Late Neoproterozoic to Early Palaeozoic palaeogeography of Avalonia: some palaeomagnetic constraints from Nuneaton, central England. Geological Magazine, **140**(6): 685–705. doi:10.1017/S001675680300832X.

Williams, H. 1979. Appalachian Orogeny in Canada. Canadian Journal of Earth Sciences, **16**(3): 792–807. doi:10.1139/e79-070.

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